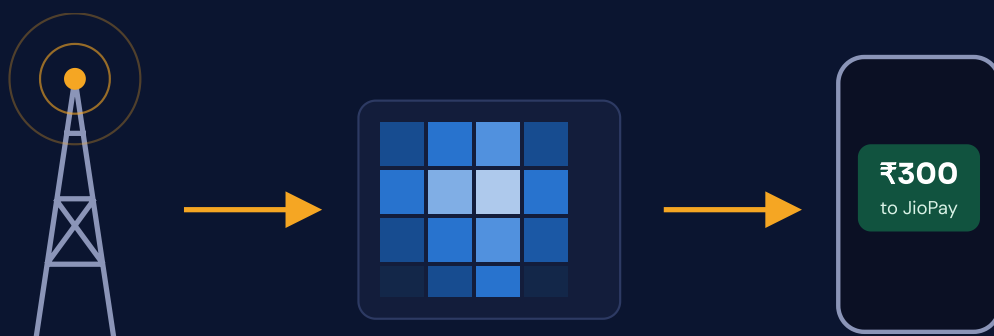


JioMausam drill book

Understand it. Explain it. Defend it. The idea, the prototype and the Indian data in plain language, then 199 jury questions with answers.

You don't need any background in science, telecom, statistics or insurance to read this. Every idea is built up from scratch with everyday examples. It takes about 40 minutes to read end to end.

Read Parts 1 and 2 in order the first time. Then drill Part 3 together until every answer comes out on its own.



1. Towers sense rain

2. A street-level rain map

3. Automatic payout

PART 1 · UNDERSTAND IT

The whole project in one minute	2	Our focus, and how realistic it is	3
Meet Ravi: the problem as a story	4	Why India can't see its weather	5
Towers that feel the rain	6	How signal becomes rain	8
Mausam Kavach: insurance, simply	10	Our prototype and its results	12

PART 2 · INDIA AND THE PRODUCT

The India layer (4 pages)	16	The Mausam Kavach phone app	20
How the money works	21	Who actually pays the ₹300?	22
Why only Reliance · risks · plan	23	Your 10 minutes with the jury	26
Dictionary of every term	27		

PART 3 · DEFEND IT

How to explain it to anyone	30	Hardening the idea: expert feedback	32
How to drill, together	37	199 cross-questions	38
Self-test	48		

The whole project in one minute

If you read only one page, read this one. Everything else in this guide is a deeper explanation of these five sentences.

- 1 **Bad weather costs poor workers money.** When heavy rain or extreme heat hits, delivery riders, street vendors and labourers can't work, and they lose that day's income. Nobody pays them back.
- 2 **India can't see weather street by street.** Our weather instruments are few and far apart, so nobody knows exactly which streets got hit and which didn't.
- 3 **Mobile towers can already "feel" the weather.** Towers talk to each other with invisible radio beams. Rain makes those beams weaker, and the weaker the beam, the harder it's raining. The network already records this.
- 4 **We turn that into a live weather map and an automatic insurance product.** The map is **JioMausam**. The insurance is **Mausam Kavach**: add it at recharge (from ₹29 a month, priced city by city) or get it free through your delivery app, and if extreme rain or heat hits where you are, ₹300 lands in your JioPay wallet within hours. No forms.
- 5 **We built a working prototype that proves it.** We used real signals from 500 tower-to-tower links and compared our rain readings with an official weather radar. They matched very closely. Then we priced the insurance on 25 years of Indian weather records.

Three names to remember

NAME	WHAT IT IS	MEANING OF THE WORD
JioMausam	The whole platform: the system that reads tower signals and draws the live weather map.	<i>Mausam</i> = weather (Hindi)
Mausam Kavach	The insurance. Basic : people buy it, ₹29–52 a month by city. Plus : delivery apps buy it for riders.	<i>Kavach</i> = shield or armour
Control Room	Our working model on real data. A second prototype shows the Kavach phone app.	Like a TV control room full of live screens

SAY IT LIKE THIS

"Every Jio tower is a rain gauge and every phone is a thermometer. We map India's weather street by street, and pay workers within hours when it stops their work."

Rain first. Riders first. Mumbai first.

JioMausam could do a lot at once: rain, heat, every city, every worker. We deliberately start narrow, with the part that is most proven and most likely to sell, and grow in phases.

Rain first

Rain from tower links is proven science, and our prototype matched radar at 0.95. Heat from phones is newer, so it comes second.

Riders first

Poor workers rarely buy insurance themselves. Delivery apps buy Kavach Plus for all their riders at once.

Mumbai first

About 12 heavy-rain days a year (Nagpur 2.4): enough events in one monsoon to test accuracy. Huge rider market.

How realistic is each part? Our honest scorecard

PART	REALISM	WHY
Measuring rain from tower links	High	Peer-reviewed since 2006; national scale in the Netherlands and Germany; our 0.95 on 500 real links.
Automatic weather insurance in India	Medium-high	Already exists: government weather-index crop insurance, and SEWA's 2023–24 heat pilots that paid automatically.
Enough microwave links in cities	Biggest unknown	Much of Jio's network is fibre, which can't sense rain. Only Jio knows the count. The pilot measures it first.
Accurate enough to pay people	Medium	We catch 68% of radar's heavy-rain events. Blending in gauges and radar must lift that.
Heat from phone batteries	Medium-low	Published research, but nobody runs it as a service yet. Phase two.
Workers buying it themselves	Low to medium	Rare without help; apps and state schemes paying is far more realistic. Phase three.
Regulator approval	Medium	Possible through IRDAI's sandbox with a licensed partner; expect a year or more.

The four phases

1 • Months 0–3	Mumbai coverage audit and shadow pilot: rain, riders, one delivery app. Payouts simulated; no money moves.
2 • Months 4–12	Real payouts in IRDAI's sandbox for one app's riders; heat module in Nagpur and Ahmedabad.
3 • Year 2	Kavach Basic at every recharge, priced city by city (₹29–52 a month).
4 • Year 3	National map for every district, with a data API for insurers, states and Reliance.

SAY IT LIKE THIS

“The science is proven, weather insurance already exists in India, and our one real unknown, link coverage, is exactly what our 90-day Mumbai pilot measures first.”

Meet Ravi

Ravi is a food-delivery rider in Mumbai. He earns about ₹800–1,000 on a good day. He gets paid per order, so no orders means no money.

A Tuesday in July, without JioMausam

At 2 pm the sky over Andheri turns black. In half an hour a cloudburst floods the roads. Restaurants stop taking orders, and riding a scooter through knee-deep water isn't safe. Ravi waits under a flyover for three hours. He makes ₹180 that day instead of ₹900.

Two kilometres away in Jogeshwari it barely drizzled. The weather report on TV just said "heavy rain in Mumbai". Nobody measured what happened on Ravi's streets, so nobody could help him even if they wanted to. No insurance company sells cover for "I lost ₹700 today", because checking every such claim would cost more than paying it.

The same Tuesday, with JioMausam

Last month his delivery app added **Mausam Kavach Plus** for all its riders, free to them. Ravi didn't have to do anything.

On Tuesday, the towers around Andheri notice their signals getting much weaker: heavy rain. The system works out that 14 mm of rain fell on Ravi's area between 2 and 5 pm. That is more than the 10 mm trigger in his policy. At 5:10 pm his phone buzzes:

MESSAGE ON RAVI'S PHONE

₹300 credited to your JioPay wallet. Mausam Kavach: 14 mm of rain was measured near you between 2:00 and 5:00 pm today. Nothing to do. Stay safe.

He didn't file anything. Nobody came to inspect. It doesn't replace his whole day's earnings, but it covers his fuel and dinner, and it arrived the same evening.

And in May, Sunita

Sunita sells cut fruit from a cart in Ahmedabad. In a heatwave, nobody shops between noon and 4 pm, and standing in 46°C heat is dangerous. She pays ₹29 a month for **Kavach Basic**. On Ahmedabad's rare 47°C days (about one a year), she gets paid in the same automatic way.

IN PLAIN WORDS

That's the whole product. Everything else in this guide explains how the towers "know" it rained, how we proved it works, and how it makes money for Reliance.

Why India can't see its own weather

Rain doesn't fall evenly: one neighbourhood floods while the next stays dry. To help people like Ravi, you need to know what happened on *each street*. India measures weather far more coarsely than that.

The three tools India uses today

TOOL	HOW IT WORKS, SIMPLY	ITS WEAKNESS
Weather radar	A big rotating dish on a tower. It sends out radio pulses that bounce off raindrops, like a bat using echoes, and works out where it's raining.	Very expensive (crores each). Only about 40 for the whole country.
Rain gauge	Basically a measuring cup left outside. Whatever water collects is the rainfall.	Measures only the exact spot it's in. Many are read once a day, not live.
Satellite	A camera in space looking down at clouds.	Sees the tops of clouds, not what actually reaches the street. Each picture covers big squares, around 10 km wide.

THINK OF IT LIKE THIS

Imagine trying to watch every street in India with only 40 CCTV cameras. You'd see the big picture, but you'd have no idea what happened on any particular lane. That's India's weather data today.

Why insurance can't help Ravi today

Insurance is simple at heart: many people each pay a small amount into a shared pot, and the few who suffer a loss get paid from it. Normal insurance asks you to **prove your loss**: you file a claim, someone checks it, there's paperwork, and you wait weeks or months.

For a car accident worth ₹50,000, that checking is worth doing. For Ravi's ₹700 lost afternoon, the checking would cost more than the payout. So nobody offers it, and millions of workers go uncovered.

~90%

of Indian workers are in the informal sector: no salary, no paid leave, no cover when weather stops work.

34 million

full-time jobs' worth of working hours India could lose to heat by 2030 (International Labour Organization).

THE PROBLEM IN ONE LINE

Nobody has a cheap, live, street-by-street record of the weather. So nobody can protect workers against it quickly.

Towers that can feel the rain

First: how mobile towers talk to each other

When you make a call, your phone talks to the nearest tower. That tower then has to pass your call on towards the rest of the network. It does this in one of two ways:

- **Fibre cable:** a thin glass cable, usually underground, that carries information as flashes of light. Rain has no effect on it.
- **Microwave link:** where there's no cable, two dishes on two towers point at each other and send a narrow, invisible radio beam across the gap, often 1 to 30 km. The word "microwave" just means the type of radio wave (the same family your microwave oven uses, but far weaker and harmless).

Clear day



Signal arrives strong

Rainy day



Raindrops soak up the beam, so it arrives weaker

What rain does to the beam

Raindrops soak up and scatter part of the beam on its way across. **More rain means a weaker signal at the other end.** The amount of weakening follows a well-known scientific formula, so you can run it backwards: from "how much weaker did the signal get?" to "how hard was it raining?"

THINK OF IT LIKE THIS

Shine a torch at a friend across the road. On a clear night they see it brightly. In heavy rain or fog, the beam looks dimmer by the time it reaches them. If you knew exactly how dim it got, you could tell how heavy the rain was.

The best part: the data already exists

To keep the network healthy, every link already records how strong its signal was when it was **sent** and when it was **received**, every few minutes, like a diary. Engineers use it to spot faults. We use the same diary to read the weather. **No new machines, no new towers.**

SAY IT LIKE THIS

"We don't build sensors. We become the sensor."

Phones as thermometers

Rain is half the story. For heat, we use the phones themselves. Every smartphone has a temperature sensor inside its battery (it's there to stop the battery overheating). A phone's battery warms up and cools down with the air around it.

One phone on its own is unreliable: it might be in a pocket, charging, or in an air-conditioned office. But **average thousands of phones in the same area** and the odd ones cancel out. What's left closely follows the real air temperature. Scientists showed this with real data from eight cities in 2013.

THINK OF IT LIKE THIS

Ask one person to guess how many sweets are in a jar and they could be way off. Ask a thousand people and average their guesses, and the answer is usually surprisingly close. This is called the "wisdom of the crowd".

BE HONEST ABOUT THIS

Our prototype proves the **rain** part on real data. Heat is our second step. The science for it is published, but we haven't built that module yet. If asked, say exactly that.

Is this real, or a crazy idea?

It's real. Scientists have been doing this in research for almost 20 years. What nobody has done is turn it into a product at India's scale.

WHEN	WHO	WHAT HAPPENED
2006	Scientists in Israel	First proved that mobile-network links can measure rain. Published in <i>Science</i> , one of the world's top journals.
2013	Scientists in the Netherlands	Made rain maps of their whole country using one phone company's links.
2013	Same Dutch group	Showed that phone battery temperatures can track city temperatures.
2016 on	Tomorrow.io, a US company	Started as a weather business using wireless signals as sensors.
2020 on	Germany and Sweden, with Ericsson	Thousands of links processed live; data and free software shared with the world.

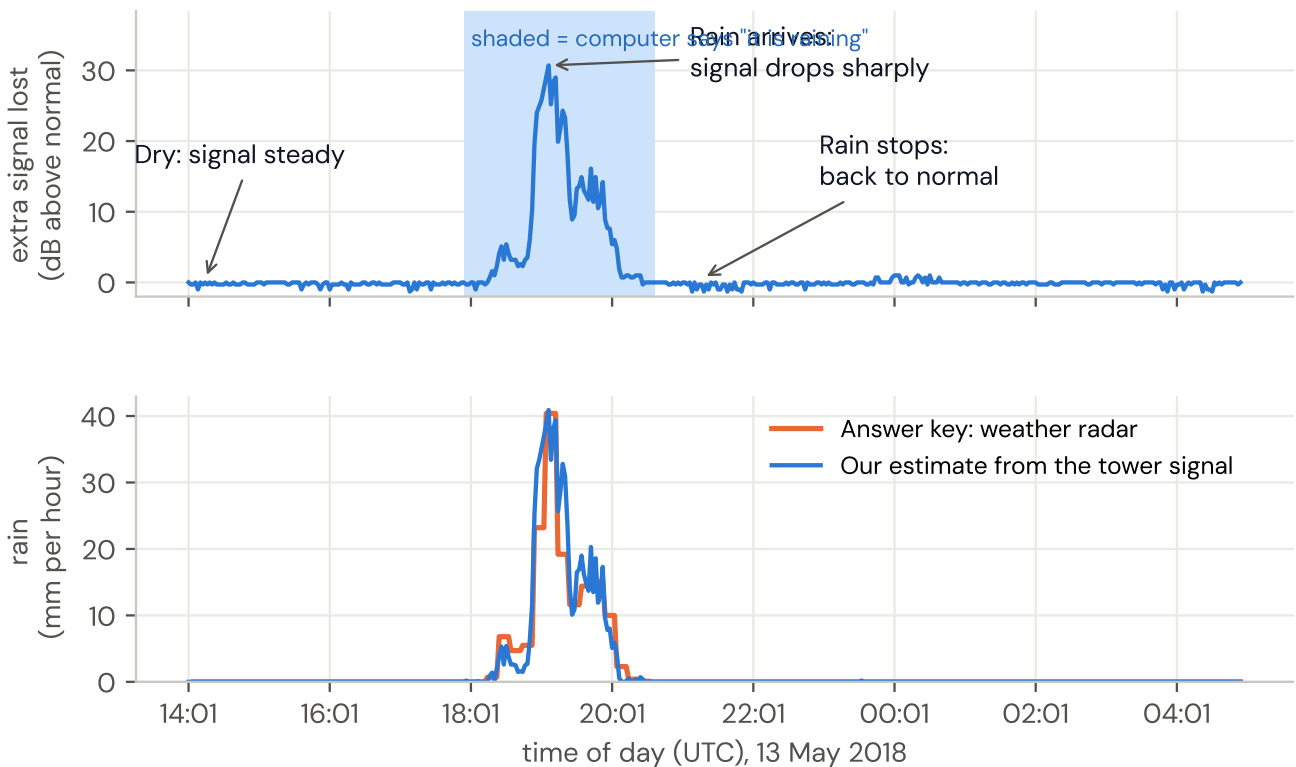
IN PLAIN WORDS

"Published in a journal" or "peer-reviewed" means other independent scientists checked the work before it was printed. It's the gold standard of "this has been verified".

How signal becomes rain, step by step

A computer program does this automatically for every link, every minute. In tech-speak this chain of steps is called a **pipeline**: data goes in one end, and rain comes out the other.

- 1 **Measure the loss.** Take “signal strength sent” minus “signal strength received”. That’s how much signal got lost on the way. It’s measured in **decibels (dB)**, a unit for signal strength. Every 3 dB lost means roughly half the signal is gone.
- 2 **Spot when it’s raining.** On a dry day the loss is steady. When rain arrives it starts to wobble and jump. The computer watches how much it has wobbled over the last hour. If the wobble is unusually large for that particular link, it says “wet”.
- 3 **Know what “normal” looks like.** Every link loses some signal even on a dry day. That normal level is called the **baseline**. Rain is only the extra loss above it. *Like knowing your normal body temperature is 37°C, so you can tell when you have a fever.*
- 4 **Remove the wet-glasses effect.** When it rains, a thin film of water forms on the dish itself and blocks a bit of signal, even in a light drizzle. This is the **wet antenna effect**. We subtract it so we don’t overcount. *Like wet spectacles making everything look blurrier than it really is.*
- 5 **Convert to rain.** A standard formula from the **ITU** (the United Nations body for telecom) turns “extra loss per kilometre” into “millimetres of rain per hour”. *Like a conversion table between inches and centimetres.*



A real example from our prototype: one 7.4 km link during a storm. Top: the signal loss jumps when rain arrives and goes back to normal when it stops. Bottom: the rain we calculated (blue) closely follows the official weather radar (orange).

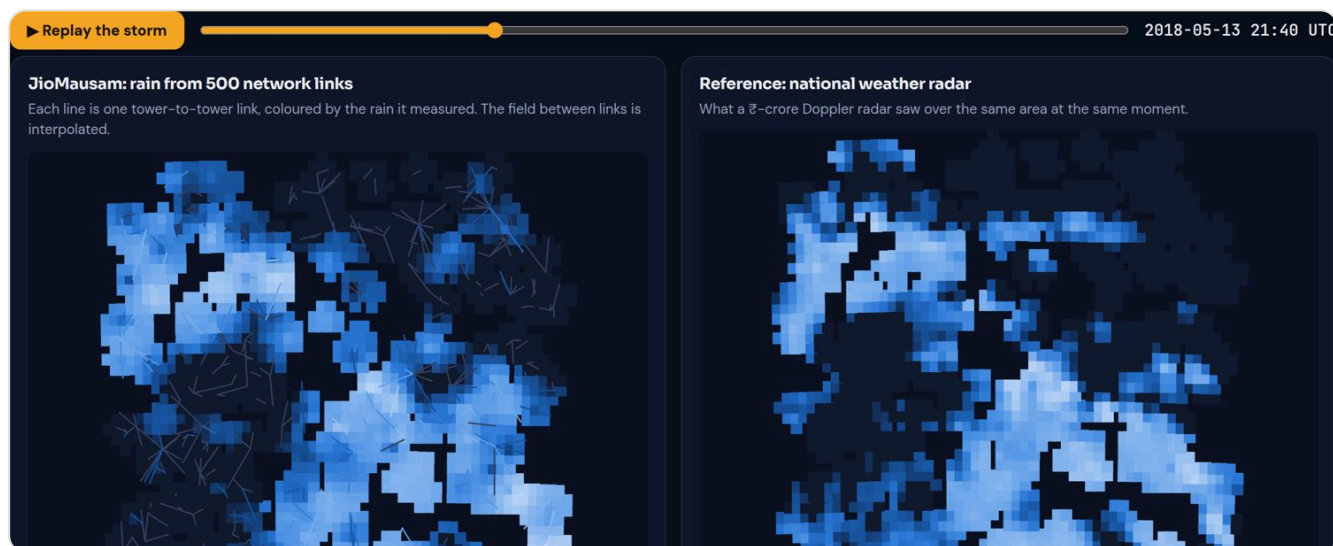
Turning hundreds of links into one map

Each link tells us how hard it rained **along its own line** between two towers. A city has hundreds of these lines criss-crossing it, which gives a web of rain readings.

To draw a map, we split the area into small squares (about 1 km in a real launch, 4 km in our prototype) and estimate each square's rain from the nearest links. This “filling in the gaps” is called **interpolation**. Our method, **inverse distance weighting**, just means “nearer links count more”. The map refreshes every 10 minutes.

THINK OF IT LIKE THIS

If it's 30°C in Pune and 34°C in a town 100 km away, your best guess for a village halfway between is about 32°C. If the village is much closer to Pune, you guess nearer 30°C. That's interpolation.



From our prototype: the same moment of a real storm. Left, our map made only from network links (each thin line is one link). Right, the official weather radar. The shapes and the heavy patches match.

How much rain is “a lot”?

Rain is measured in **millimetres (mm)**: if 1 mm of rain fell, it would make a layer of water 1 mm deep on flat ground. “mm per hour” tells you how fast it's falling.

RAIN RATE	WHAT IT FEELS LIKE
under 2.5 mm/hour	Light rain. You might skip the umbrella.
2.5 to 7.6 mm/hour	Moderate. Puddles form, and two-wheelers slow down.
over 7.6 mm/hour	Heavy. Roads start to flood and deliveries stop.
100 mm/hour or more	What IMD calls a cloudburst. Flash floods.

Our Kavach trigger of **10 mm within 3 hours** means a few hours of steady heavy rain: the kind that actually stops outdoor work.

Insurance, simply

The basic words

Policy	The agreement: what you're covered for and what you'll get.
Premium	What you pay for the cover. For Kavach, ₹29 a month.
Payout	The money you receive when the covered event happens. For Kavach, ₹300.
Claim	In normal insurance, your request to be paid, with proof. Kavach doesn't need one.
Insurer / underwriter	The licensed company that takes the risk and pays out. Not Jio: a partner insurance company.

Normal insurance vs parametric insurance

Normal insurance ("indemnity")



Parametric insurance (Mausam Kavach)

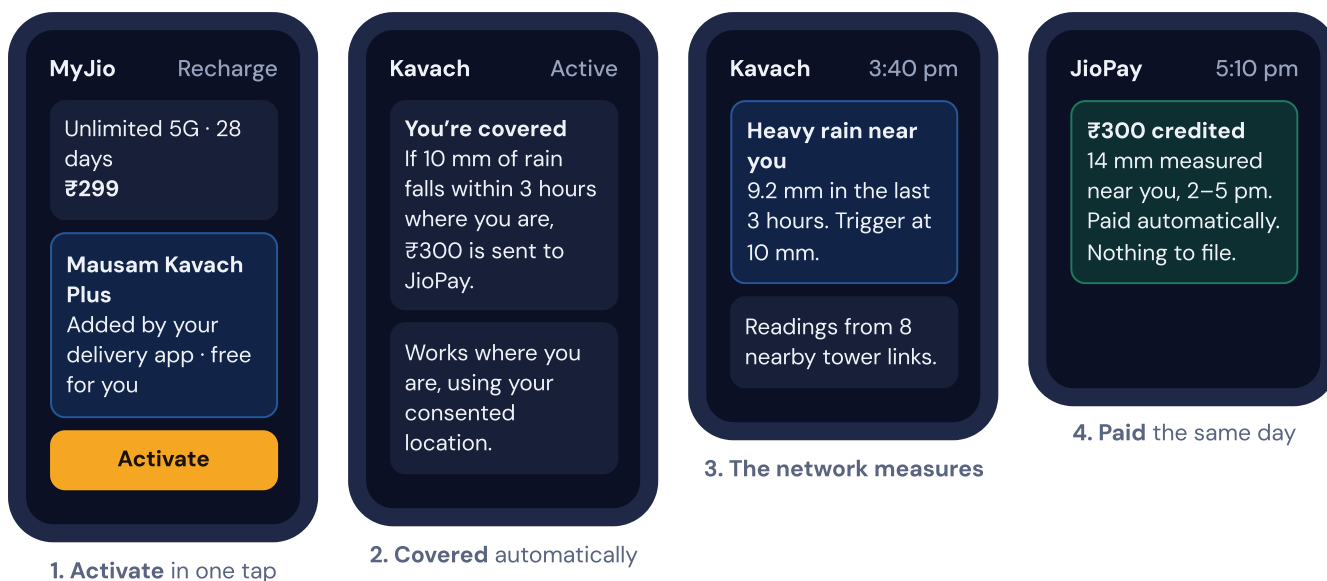


Parametric insurance doesn't pay for your *proven loss*. It pays when a **measured event** happens, according to a rule agreed in advance. That rule is called the **trigger**. Kavach Plus pays on any work-stopping day, such as "10 mm of rain within 3 hours in your 1 km square". Kavach Basic pays only on your city's rare extreme days, like 47°C in Ahmedabad. Since nobody has to check anything, a tiny ₹300 payout becomes affordable to process.

THINK OF IT LIKE THIS

Some travel insurance pays you automatically if your flight is more than 2 hours late. The airline's official records prove the delay, so you don't send any documents. Kavach works the same way, except the official record is the weather our network measured.

How a payout works, from the customer's side



How a price is worked out: a real example from Ahmedabad

- 25 years of IMD records show Ahmedabad reaches 47°C or more on about **0.7 days a year** (0.64 once payouts are capped at 5 days).
- Expected payout per customer: $0.64 \times ₹300 = ₹192$ a year.
- We want about 60% of premiums paid back as payouts (the **loss ratio**). So the premium is $₹192 \div 0.6 = ₹320$ a year, about **₹27 a month**. We charge **₹29**.

Mumbai is different. It gets about 12 heavy-rain days a year, so covering all of them would cost about ₹208 a month. That's why Kavach Basic in Mumbai covers only "extremely heavy" days (about 1 a year, ₹45 a month), and delivery apps buy Kavach Plus for riders who need cover on every work-stopping day.

Three more words you'll hear

Basis risk	The chance that the rule and real life don't match. Example: Ravi's square got 9 mm (no payout), but his street flooded badly anyway. Or the reverse. We reduce it with small squares, many data sources and a review option for disputes.
Waiting period	Cover starts 7 days after you buy, so nobody can buy it only when a storm is forecast.
Cap	A maximum number of payout days per year, so costs stay predictable.

What we built, and what we tested

A **prototype** is a working early version that shows an idea really works. It isn't the finished product. Think of a model car that actually drives, before the factory is built.

Ours is a website called the **JioMausam Control Room**. It doesn't use made-up numbers: it runs our rain method on **real signal data from 500 microwave links** of a real mobile network, recorded every minute for 11 days in May 2018.

Why the data is from Germany, not India

Phone companies don't publish their link data. Scientists have released one real dataset for research, and it happens to be from Germany (the exact tower locations are shifted for privacy). The physics of rain and radio waves is the same everywhere, and Indian monsoon rain is heavier, which makes the signal even easier to read. Testing on Indian links is exactly what the pilot is for. For Indian evidence on rain patterns and pricing, see the India layer chapter.

How we checked if it works

The same dataset includes readings from Germany's official **weather radar**. We used the radar as our **answer key**: for every link and every hour, we compared "what we calculated" with "what the radar saw".

CALIBRATION

We used the first 3 days to adjust our numbers once, by a single factor ($\times 1.17$), because our method read slightly low. *Like adjusting a kitchen scale that reads a little light.*

HELD-OUT DAYS

We then tested on the other 8 days, which the method had never seen. *Like testing a student on exam questions they haven't practised, so you know they really understand.*

BE HONEST ABOUT THIS

The radar is not perfect either. It's the best "answer key" available, but it has its own errors. When we "disagree" with it, sometimes the radar is the one that's wrong.

The prototype has four screens ("tabs")

Live rain map	Replays a real storm, every 10 minutes: our map next to the radar.
How one link sees rain	The five-step method on one real link (the chart on page 7).
Accuracy	The report card: how close we were to the radar.
Kavach payouts	The insurance in action: 1,000 pretend customers, real storm data, automatic payouts.

Reading our results, one number at a time

First, what “correlation” means

Correlation (written r) is a score from 0 to 1 for how closely two sets of readings rise and fall together. 0 means no relationship at all. 1 means they move perfectly in step.

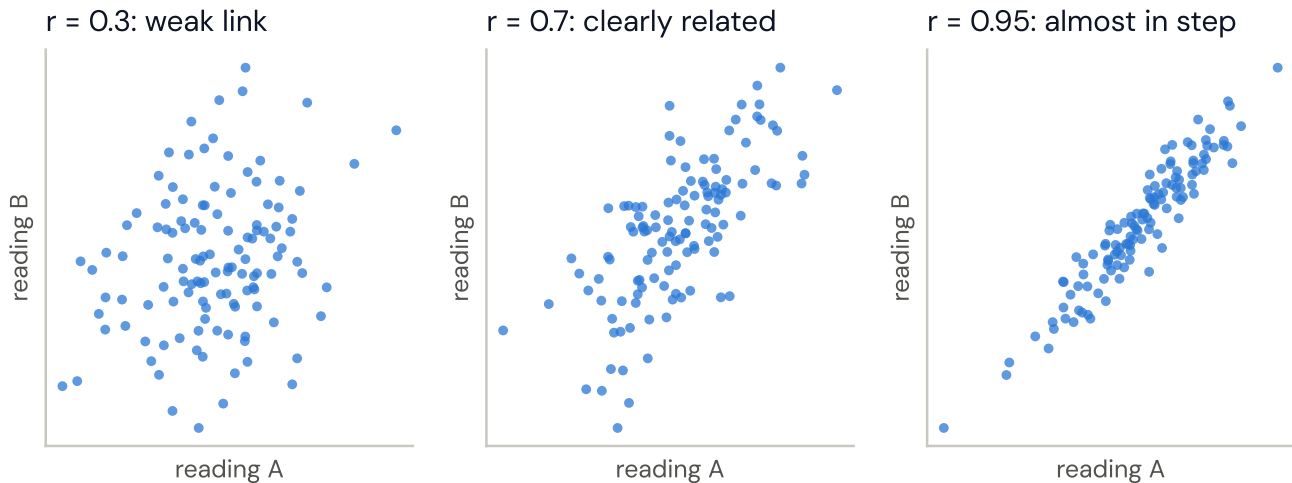


Illustration with made-up dots, just to show what different scores look like. Each dot is one moment: reading A across, reading B up.

0.95

Correlation with radar for the whole network’s average rain, hour by hour.

When the radar said “more rain”, we said “more rain” too, almost perfectly. Say: “Network-wide, we match radar at 0.95.”

0.88

Correlation for a typical single link. Even one pair of towers on its own tracks the radar very well. 91% of links scored above 0.7.

91%

Rainy days caught (the detection rate): of all the days the radar said a link’s path got rain, we said so too 91% of the time. **4.6% false alarms**: of the days we said “rain”, only 4.6% the radar didn’t agree. Like a smoke alarm that rings for 91 of 100 real fires and almost never rings for burnt toast.

94.6%

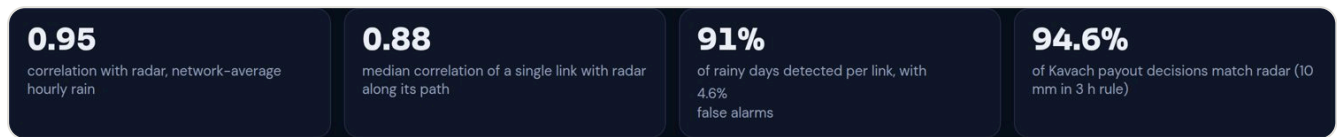
Same payout decision as radar. For every 4 km square on every test day, we asked “would Kavach pay?” using our data and using radar data. They gave the same answer 94.6% of the time.

BE HONEST ABOUT THIS

94.6% sounds great, but it’s high partly because most days are dry, and saying “no payout” on a dry day is easy. (A forecaster in a desert who always says “no rain” is right most days.) The tougher number: of the heavy-rain events the radar saw, we caught **68%**. Also, our rain totals came out about **18% lower** than radar. Both get better in the pilot by adding more links, rain gauges and radar data. Judges respect you more for saying this yourself.

A tour of the Control Room, part 1

Open the link to the prototype in a browser on a laptop. At the top you'll always see four headline numbers. Under them are the four tabs.



The four headline numbers. They're explained on page 12.

Tab 1 • Live rain map



- **What you see:** two maps of the same area at the same moment. Left is ours, right is the radar. Darker blue is lighter rain; brighter blue is heavier.
- **What to click:** "Replay the storm". The real storm of 13–14 May 2018 plays in 10-minute steps. You can drag the slider to any moment.
- **What to say:** "Left is our network, right is a radar worth crores. Same storm, and we didn't need a radar."
- **The boxes underneath:** average rain right now (ours vs radar), how many of the 500 links are currently sensing rain, and how much of the area is getting rain.

A tour of the Control Room, part 2

Tab 2 - How one link sees rain



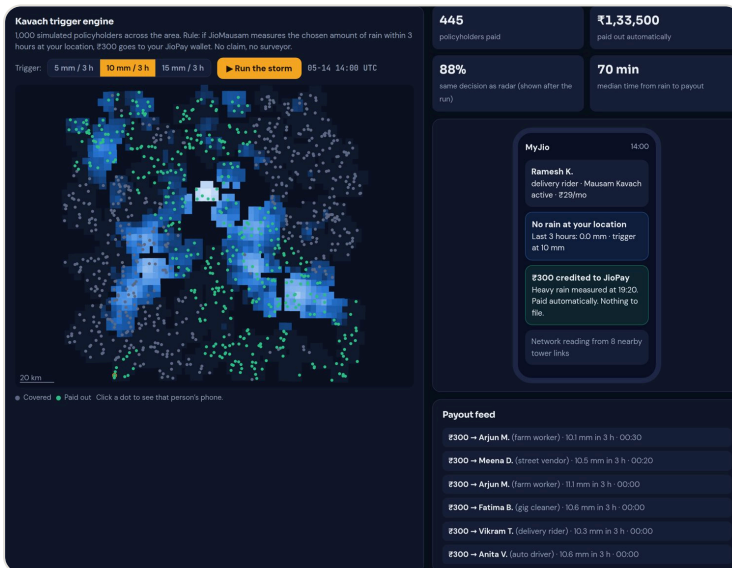
The five steps from page 7 shown on one real link. The top chart is signal loss (blue shaded bands are where the computer decided “raining”). The bottom chart is our rain vs the radar. Hover over a chart to read exact values.

Tab 3 - Accuracy



The report card. Top left: 11 days of average rain, ours and radar’s (grey band = calibration days). Top right: one dot per link per hour; dots near the diagonal line mean we agreed. Bottom: how much of the radar’s rain we caught, with false alarms.

Tab 4 - Kavach payouts



- **The dots** are 1,000 pretend customers spread across the area. Grey = covered, green = paid.
- **The trigger buttons** (5, 10 or 15 mm in 3 hours) change the rule. A lower rule means more payouts, and more chance of paying when it wasn’t needed.
- **“Run the storm”** plays the storm. Watch dots turn green and the payout feed fill up.
- **Click any dot** to see that person’s phone, as in Ravi’s story.
- **The result:** with the 10 mm rule, 445 of 1,000 people were paid ₹1,33,500 in total, about 70 minutes after their rain started.

Why our tower test is German, and where our Indian evidence comes from

Why we couldn't test on Indian towers

The method needs one specific kind of data: how strong every tower-to-tower radio signal was when it was sent and when it arrived, minute by minute. Only the phone company has these records. Jio, Airtel and Vi keep them internal because they reveal how their networks are built. No open dataset of this kind exists for India.

So we proved the method on the best open dataset in the world (500 real links from Germany), and used Indian weather records to answer the two questions that don't need tower data:

1. Is Indian rain really hyperlocal?

If rain fell evenly across a city, nobody would need street-level data.

2. What should Kavach cost in each city?

How often does each city actually hit a payout trigger?

Where the Indian data comes from

Stations	Mumbai Santacruz (airport), Mumbai Colaba (south Mumbai, about 20 km away), Nagpur and Ahmedabad airports. All official IMD stations.
Years	2000–2024: 25 years of daily readings.
What's measured	Daily rainfall (24 hours to 8:30 am) and daily maximum temperature.
Source	IMD's station reports as archived by NOAA, the US weather agency, in its "Global Summary of the Day", which collects official station reports from every country. We reached it through NOAA's open data on Amazon's cloud.

BE HONEST ABOUT THIS

The archive is good for temperatures and for counting rain days, but it can miss part of a single extreme day: it shows Mumbai's 26 July 2005 deluge as 461 mm, while IMD's official figure is 944 mm. So we use it to count how often thresholds are crossed, not for record totals. Its 25-year averages match IMD's published normals: Mumbai about 2,666 mm a year, Nagpur 1,047 mm, Ahmedabad 859 mm.

SAY IT LIKE THIS

"No Indian operator publishes link data, so we proved the method on the best open dataset in the world and priced it on 25 years of IMD records. Give us Jio's link data for Mumbai and we'll prove it on Indian rain in one monsoon."

Is Indian rain really hyperlocal? Yes.

944 mm

Santacruz, 26 July 2005

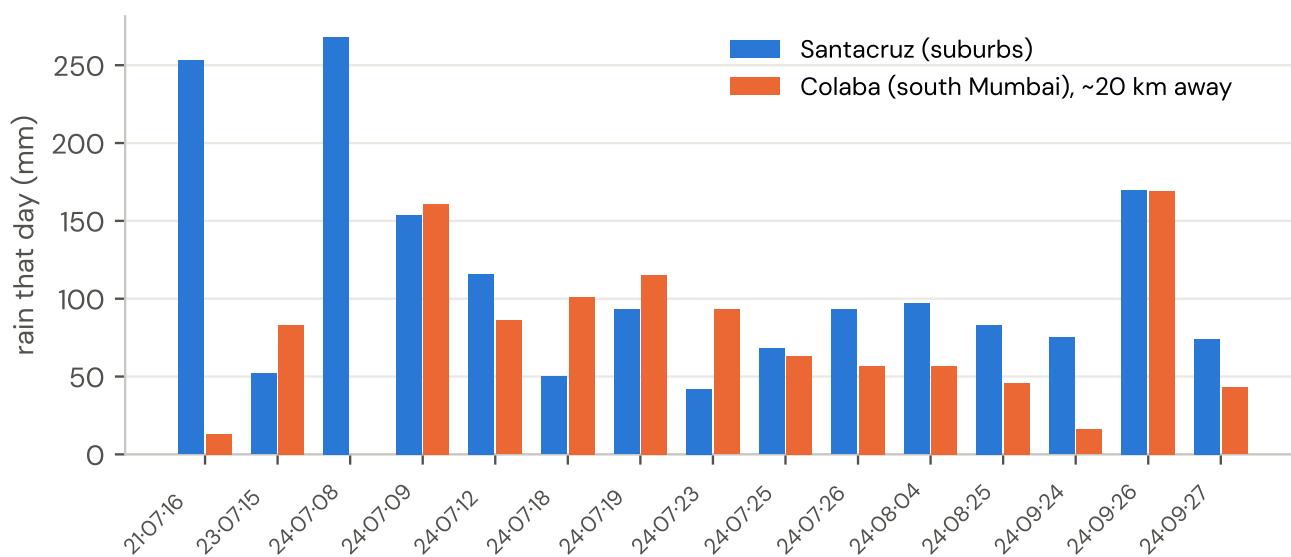
73 mm

Colaba, about 20 km away, same day

Mumbai's worst flood day shows it perfectly. Santacruz in the suburbs recorded **944 mm** of rain in 24 hours, among the heaviest daily totals ever measured in India. Colaba in south Mumbai, about 20 km away, recorded **73 mm**. Same city, same day, about thirteen times different.

Source: IMD's official figures, widely reported.

And it isn't a one-off



Every heavy-rain monsoon day (64.5 mm or more at either station) from 2019 to 2024 where both stations sent a valid 24-hour report: 15 days. On **5 of the 15 (33%)**, one station got less than half the other's rain.

IN PLAIN WORDS

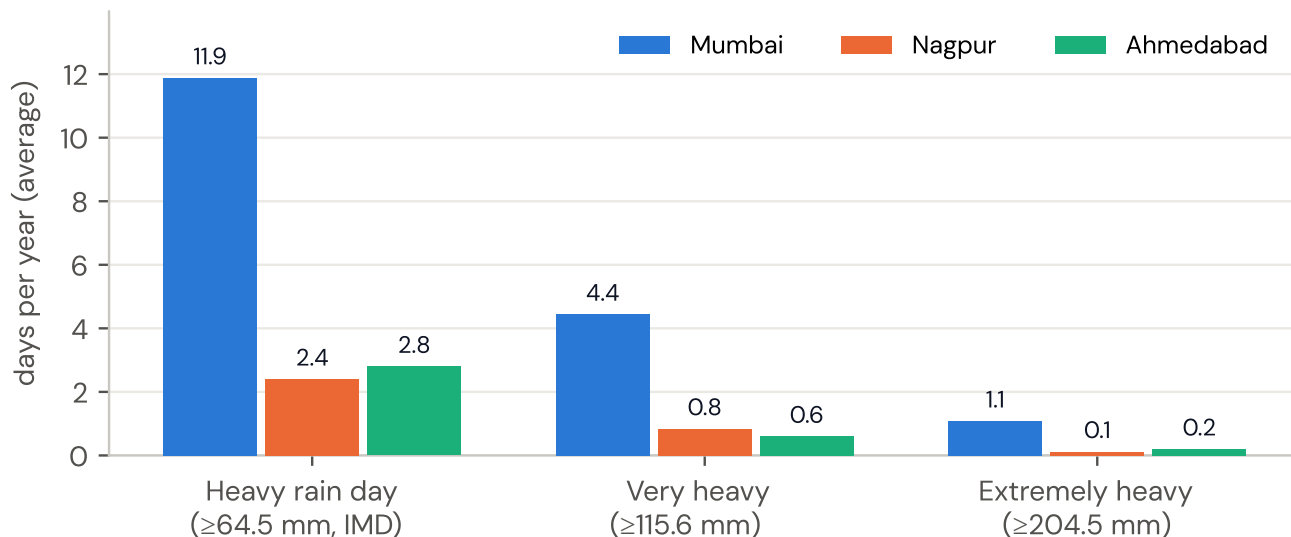
A single weather station, or a city-wide forecast, regularly gets a neighbourhood's rain badly wrong. A worker 20 km from the gauge can be flooded out while the official reading says "moderate". That's the gap tower links fill: a city has hundreds of links, not two gauges.

SAY IT LIKE THIS

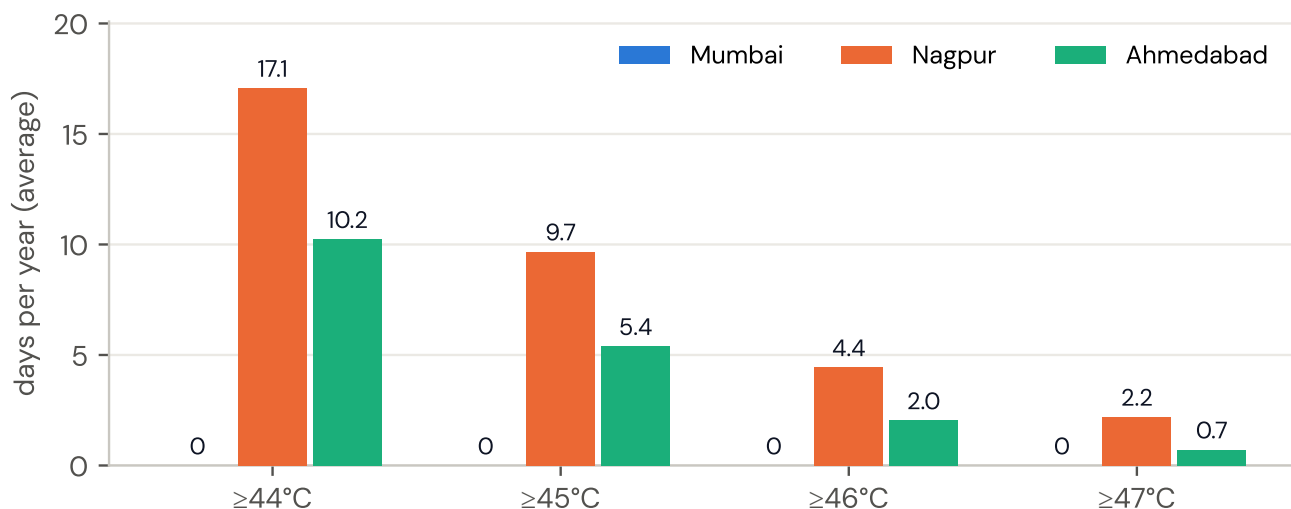
"On a third of Mumbai's heavy-rain days, a station 20 kilometres away got less than half the rain. Two gauges can't see a city. Hundreds of tower links can."

How often would Kavach pay in each city?

For each city and each year from 2000 to 2024 we counted the days that crossed each threshold, then averaged. Rain uses IMD's own categories: **heavy** (64.5 mm or more in a day), **very heavy** (115.6 mm+) and **extremely heavy** (204.5 mm+).



Rain: average days per year in each IMD category. Mumbai is in a different league, with about 12 heavy-rain days a year.



Heat: average days per year at or above each maximum temperature. Nagpur and Ahmedabad face heat; Mumbai never reaches 44°C.

WHAT THIS TELLS US

Every city has a different main danger. Mumbai's is rain; Nagpur's and Ahmedabad's is heat. And how often a trigger fires varies hugely: a 45°C trigger fires about 10 times a year in Nagpur and never in Mumbai. One rule and one price for all of India can't work.

City-by-city pricing, and two plans

The same rule for every city: ₹300 per payout, at most 5 payout days a year, and about 60% of premiums paid back. **Monthly price = average payout days × ₹300 ÷ 0.6 ÷ 12.**

CITY	KAVACH BASIC PAYS ON	HOW OFTEN	BASIC	KAVACH PLUS (BOUGHT BY PLATFORMS)
Mumbai	Extremely heavy rain (204.5 mm+ in a day)	1.1 a year	₹45	Any heavy-rain day or 10 mm in 3 h: about ₹208 a rider a month
Nagpur	Heat of 47°C or more	2.2 a year	₹52	45°C+ days: about ₹168 a worker a month
Ahmedabad	Heat of 47°C or more	0.7 a year	₹29	45°C+ days: about ₹100 a worker a month

Two plans, two kinds of customer

Kavach Basic: for individuals

Bought at recharge for ₹29–52 a month depending on the city. Pays on the city's rarest, most dangerous days, about one or two a year. Cheap and simple: a true safety net.

Kavach Plus: for platforms

Bought by delivery and gig apps for their workers. Pays on any work-stopping day, up to 5 a year. In Mumbai about ₹208 a rider a month (about ₹7 a day), so 50,000 riders cost about ₹12.5 crore a year.

WHAT CHANGED, AND WHY THAT'S A STRENGTH

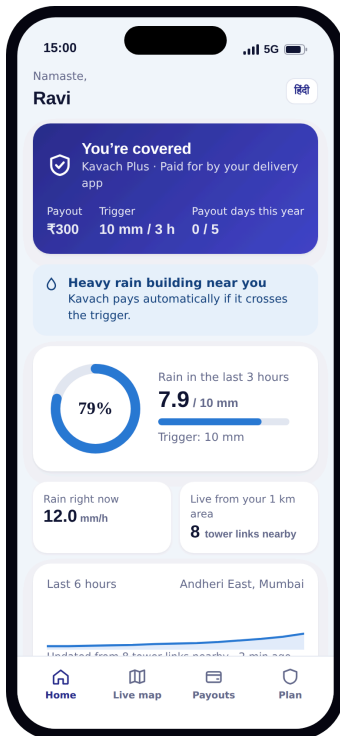
Our first pitch said "₹29 a month, everywhere". Indian data showed that can't work: Mumbai has about 12 heavy-rain days a year. So we did what a real insurer does and priced each city on its own history. Judges respect a team that tested its own assumption and changed its mind on evidence.

SAY IT LIKE THIS

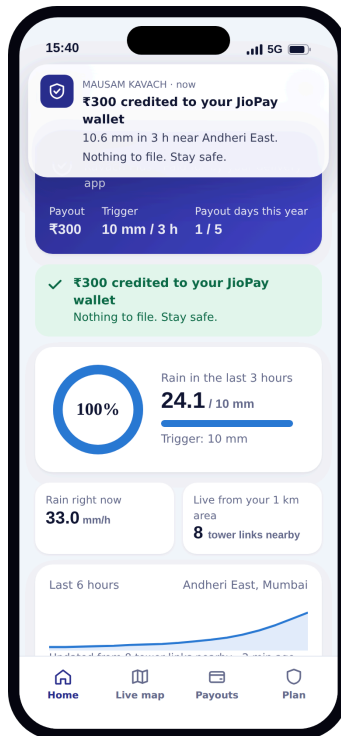
"Indian weather changed our pricing. Mumbai's risk is rain; Nagpur's and Ahmedabad's is heat. So Kavach Basic covers each city's extreme days for ₹29 to ₹52 a month, and delivery apps buy Kavach Plus for their riders."

The Mausam Kavach app, screen by screen

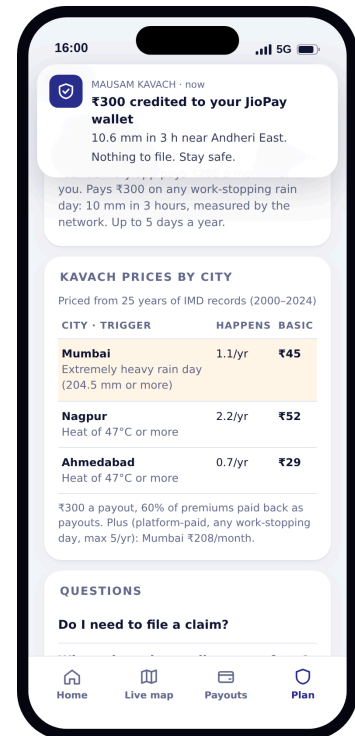
Our second prototype shows what a worker actually sees. It opens in any browser and looks like a phone; on a real phone it fills the screen.



Home. “You’re covered”, the plan, and a ring that fills as rain or heat builds towards the trigger.



The payout. The moment the network crosses the trigger, the notification drops in: ₹300 to JioPay, with the reading.



Plan. City prices set from 25 years of IMD records, and simple answers to common questions.

How to demo it

1. Open the app (link or standalone file) on a laptop. The left panel is your presenter control; the phone is on the right.
2. Pick a person: **Ravi** (Mumbai rider, Plus), **Sunita** (Ahmedabad fruit vendor, Basic ₹29) or **Ramesh** (Nagpur construction worker, Basic ₹52).
3. Press **Play the storm** (or the heatwave). The ring fills; when it crosses the trigger, the ₹300 notification drops in. Tap it to see the reading.
4. Switch to **हिंदी** to show it works in Hindi. Open **Payouts** to show real past dates when Kavach would have paid, from IMD records.

BE HONEST ABOUT THIS

The live storm and heatwave are demo scenarios. The prices and the past payout dates are real IMD data. If asked: “The scenario is a demo; the pricing and the history are real Indian data.”

How the money works

A business idea needs to explain who pays, how much, and why it's cheap to deliver. JioMausam earns in three ways.

WAY TO EARN	WHO PAYS	BUSINESS SHORTHAND
1. Selling Kavach	Workers buy it at recharge. Or companies (like delivery apps) buy it for their workers.	B2C = business to consumer. B2B2C = a company buys it for its people.
2. Selling the weather data	Insurance companies, state disaster departments, city flood teams, farm and logistics companies pay to use our live map.	B2B = business to business. B2G = to government. API = how their computers plug into our data.
3. Reliance uses it itself	JioMart stocks better for rainy days, Reliance New Energy predicts solar power better, Jio predicts weather problems on its own network.	Customer zero = the company is its own first customer.

The famous ₹174 crore, step by step

- 1 Jio has about **500 million** (50 crore) users.
- 2 Suppose just **1%** buy Kavach. That's **5 million** (50 lakh) people.
- 3 Each pays ₹29 × 12 = **₹348 a year**.
- 4 50 lakh × ₹348 = **₹174 crore a year** in premiums. About 60% (₹104 crore) goes back to people as payouts. The rest pays the insurer, Jio's fees and running costs.

IN PLAIN WORDS: WHY THIS WORKS WHEN TINY INSURANCE USUALLY FAILS

Small insurance usually fails for two reasons: finding customers is expensive (agents, ads) and checking claims is expensive. Here, customers are found inside a recharge they already do every month, which costs almost nothing. In business terms, the **customer acquisition cost (CAC)** is near zero. And payouts are automatic, so there are no claims to check.

BE HONEST ABOUT THIS

These are illustrative numbers to show the size of the opportunity, not a forecast. ₹29 is the cheapest city price (Basic is ₹29–52), so ₹174 crore is the cautious case. Jio doesn't take the insurance risk itself: a partner insurer does, and Jio earns a **distribution fee** (for selling it) and a **data licence fee** (for providing the trigger data).

Who actually pays the ₹300?

Not Reliance. Insurance is a shared pot: many people each pay a little, and the few who have a bad day are paid from it. Reliance never pays out of its own pocket.

The pot, with real numbers (Ahmedabad, 1 lakh Kavach Basic customers)

Premiums in: $1,00,000 \times ₹29 \times 12 \text{ months}$	₹3.48 crore a year
Payouts out: $47^\circ\text{C on about } 0.64 \text{ days a year} \times ₹300 \times 1,00,000$	₹1.92 crore a year
Left over for the insurer, reinsurance and Jio's fees	₹1.56 crore a year

Who does what

- **Worker or delivery app:** pays the premium.
- **Insurer** (a JFS partner): holds the pot, carries the risk, pays out.
- **Jio:** sells it and supplies the readings. Earns fees, carries no risk.

Why Reliance wants it

- Fee income on every policy.
- Kavach customers stay on Jio.
- The weather data sells separately.
- JioMart, New Energy and Jio use it.

“Why would someone sitting in AC buy it?”

They wouldn't, and that's by design. Kavach is for people whose income stops when the weather turns: riders, vendors, labourers. Eligibility is checked through the delivery app or **e-Shram**, the government's register of unorganised workers (about 30 crore sign-ups). Everyone else gets something different:

WHO	WHAT THEY GET	WHO PAYS
Outdoor workers	Kavach payouts	The delivery app (Plus) or the worker (Basic)
Everyone else, including office workers	Free street-level rain and flood alerts in MyJio	Nobody; it keeps them on Jio
Businesses and government	The live weather data	They do

SAY IT LIKE THIS

“We don't give money away. Workers pool small premiums, and the network decides automatically who had the bad day.”

Five puzzle pieces that only Reliance holds

To make JioMausam work, you need five things at the same time. Most companies have one or two. Reliance has all five under one roof.

PIECE	WHAT'S NEEDED	RELIANCE ALREADY HAS
Sense	Towers everywhere people live, to measure the weather	The Jio network: lakhs of sites, every district
Reach	A way to reach hundreds of millions of people every month	500M+ Jio users who recharge, and the MyJio app
Pay	A way to send ₹300 instantly and cheaply	JioPay and JioFinance
Insure	A licensed insurance partner	Jio Financial Services and its insurance partners
Use	Businesses that benefit from weather data	JioMart, Reliance Retail, Reliance New Energy, Jio itself

Two business words for this

MOAT

Like the water ditch around a castle: something that makes it hard for competitors to copy you. Our moat is that nobody else holds all five pieces.

FLYWHEEL

A cycle that speeds itself up. More users means more phones and sites, so better maps, so fairer payouts, so more users.

Who else is out there?

- **IMD (India Meteorological Department):** the government weather office. It has radars and forecasts, but not street-level detail, and it doesn't sell insurance.
- **Private weather companies:** sell forecasts, but don't own a sensor network in every Indian street.
- **Insurance companies:** some test weather-triggered products, but lack street-level trigger data and cheap ways to reach workers.
- **Other telecom companies (Airtel, Vi):** could sense rain too, but don't have payments, insurance, retail and energy all together.

The weak spots, and our answers

Every idea has weak spots. Judges will look for them. Knowing them before they ask makes you look prepared, not weak.

THE WORRY	OUR ANSWER, IN SIMPLE WORDS
“Jio mostly uses fibre cables, not microwave beams.” Fibre isn’t affected by rain, so it gives no rain signal.	True for much of the network. But many towers, especially rural, hilly and newer ones, still use microwave links. We also add other signals (phone and tower data) and official radar and gauges. The first job of the pilot is to measure exactly how much coverage we get.
“What if someone loses money but doesn’t get paid?” This is basis risk (page 10).	Small squares, several data sources, rules set for each area, and a review option for complaints. Even before adding other sources, our prototype agreed with radar on 94.6% of payout decisions.
“Isn’t using phone data a privacy problem?”	The rain part uses tower equipment data, not personal data. The heat part only uses phones whose owners agree, combined by area so no single person is visible. This follows India’s data protection law, the DPDP Act 2023 .
“Who allows you to sell insurance?”	The insurance regulator, IRDAI . It runs a regulatory sandbox : a safe trial zone where new products are tested with a limited number of customers. We start there with a licensed partner.
“Will poor workers trust it?”	Delivery apps can buy it for their riders, so it’s free to them. Messages go out in local languages on WhatsApp. And the first automatic payout is the best advertisement.
“Your rain numbers read low.”	About 18% low in the prototype. It’s a known effect of the simple method, fixed by regularly comparing with rain gauges (calibration).
“People will buy it only when a storm is coming.”	A 7-day waiting period, and monthly auto-renewal with the recharge.

IF YOU DON’T KNOW AN ANSWER, SAY THIS

“We haven’t measured that yet. It’s exactly what our 90-day pilot is designed to find out, and here’s what we’d measure.”

What happens next if Reliance says yes

A **pilot** is a small real-world trial before going big, like opening one shop before opening a hundred. A **roadmap** is the plan of steps and timings. Ours has four phases.

WHEN	PHASE	WHAT HAPPENS, SIMPLY
Months 0–3	1 • Coverage audit + shadow pilot	Mumbai: rain, riders, one delivery app. Count the usable links in every square, check our map against IMD's instruments and the city's rain gauges, and compute every payout in shadow mode. No money moves yet.
Months 4–12	2 • Sandbox + heat	Real Kavach Plus payouts for one app's riders inside IRDAI's sandbox. Build the heat module in Nagpur and Ahmedabad, checked against IMD stations.
Year 2	3 • Kavach Basic	Anyone can add it at recharge, priced city by city (₹29–52 a month). More delivery apps join.
Year 3	4 • National map	Every district. Sell the weather data to insurers, states and businesses. Offer the method to telecoms in other monsoon countries.

Why Mumbai first

- **Enough storms to test:** about 12 heavy-rain days a year, against about 2.4 in Nagpur. One monsoon gives a dozen events to measure accuracy on.
- **Riders are there:** one of India's biggest delivery-app markets.
- **Our best Indian evidence is there:** 944 mm vs 73 mm in 2005, and the 1-in-3 finding.

How we'll know the pilot worked

- Our rain readings match IMD's at a correlation of **0.8 or more** (the prototype reached 0.88 per link).
- We catch **at least 80%** of heavy-rain events, with fewer than 15% false triggers.
- Money reaches riders' wallets within **6 hours** of the trigger.

What we'd ask Reliance for

- Jio's link data for Mumbai, handled by Jio's own team.
- One insurance partner through Jio Financial Services, and an introduction to one delivery app.
- A small team of about five people. **No new hardware.**

Your 10 minutes, second by second

TIME	WHAT HAPPENS	WHAT YOU DO
0:00–2:54	Pitch video plays	Join when your team is called, rename yourself TeamName_YourName, turn your camera on, and get the deck ready to share.
3:00–6:00	Your 3-minute presentation	9 slides, with the script in the speaker notes. Present it; don't read it.
6:00–10:00	4 minutes of Q&A	Expect 5–7 questions. One-sentence answer, one number, then stop.

The 9 slides

0:00	Cold open	"Three things: proof, where we start, our ask."
0:12	Our network vs a radar	500 real links, zero hardware. Optional: switch to the live prototype.
0:40	Tested on unseen days	0.95 network-wide, 0.88 per link, 91% of rainy days.
1:00	Where to start	Mumbai: about 12 heavy-rain days a year, so rain and riders first. Heat is phase two.
1:25	What a worker sees	Ravi's ring fills, ₹300 lands, nothing to file.
1:50	Riders first, then everyone	₹12.5 Cr of premiums per 50,000 riders; Jio keeps 15–25% as fees and carries no risk. Basic later.
2:15	Every layer, and the ask	90-day Mumbai coverage audit in shadow mode: Jio's link logs, one JFS partner, one delivery app, five people.
2:35	Four phases	Coverage audit, then sandbox and heat, then Basic at every recharge, then a national map.
2:52	Close	"India will adapt by listening to what it already has."

Tech checklist for the night before

- **Animations:** the PowerPoint uses Morph transitions and entrance animations. They play when you share your screen from PowerPoint 2019 or Microsoft 365 in Slide Show. If the platform makes you *upload* slides, it may turn them into still images, so upload the PDF version, which looks the same without the motion.
- **Prototypes offline:** download the two standalone HTML files and open them from your laptop, so they work even if a link or login fails.
- **Links:** the artifact links are private to your account. Share them from their Share menu, or use the standalone files.
- Charged laptop, strong internet plus a phone hotspot as backup, separate quiet rooms, cameras at eye level.

Every technical word, in plain language

WEATHER AND SCIENCE

Attenuation · The weakening of a signal on its way. Rain causes attenuation. “Signal loss” means the same.

Cloudburst · Extremely intense rain over a small area, about 100 mm or more in an hour (IMD’s definition).

Doppler weather radar · A big rotating dish that bounces radio pulses off raindrops to see rain and its movement. India has about 40.

Heat index · How hot it actually feels, combining temperature and humidity. 40°C in humid Mumbai can feel worse than in dry Jaipur.

Heatwave · A spell of abnormally high temperatures. IMD declares it based on local normal temperatures.

Hyperlocal · At a very small scale: street or neighbourhood, not city or state.

IMD · India Meteorological Department, the government’s weather office.

Mission Mausam · A government programme launched in 2024 (about ₹2,000 crore) to improve India’s weather observation and forecasting.

mm of rain · Depth of rainwater on flat ground. “mm per hour” is how fast it’s falling.

Peer-reviewed · Checked by independent scientists before publication. The gold standard of trust in research.

Rain gauge · A container that collects rain to measure it at one spot.

Satellite data · Pictures and measurements from space. Good for big patterns, not street detail.

Urban heat island · Cities, full of concrete, get hotter than leafy or rural areas nearby.

MOBILE NETWORKS

Antenna / dish · The device on a tower that sends and receives radio signals.

Backhaul · The connections that carry traffic from a tower back into the main network, by fibre cable or microwave link.

Base station / cell site · A mobile tower and its equipment.

CML (commercial microwave link) · The scientists’ name for a microwave link run by a phone company. Our “rain gauges”.

Decibel (dB) · A unit for signal strength. Losing 3 dB means roughly half the signal is gone; 10 dB means 90% gone.

E-band · Very high-frequency microwave links (70–80 GHz), used for fast connections. Very sensitive to rain, which is good for us.

Fibre (optical fibre) · Thin glass cable carrying data as light. Unaffected by rain.

Frequency / GHz · How fast a radio wave vibrates. GHz means billions of times per second. Our links run from 6 to 39 GHz. Higher frequencies are more affected by rain.

Microwave link · A narrow radio beam between two dishes on two towers, often 1–30 km apart.

RSL / TSL · Received Signal Level and Transmitted Signal Level: how strong the signal was when it arrived and when it was sent. The difference is the loss.

Rain fade · When rain weakens a link so much that the connection degrades. A problem for Jio, and a signal for us.

Performance logs · The network’s automatic diary of how each piece of equipment is doing. Where our raw data comes from.

4G / 5G · Generations of mobile technology. 5G is the newest in India.

ISAC · “Integrated sensing and communication”: the research idea, expected in 6G, that networks should also sense their surroundings. JioMausam is an early, practical version of it.

Data, numbers and insurance words

DATA AND NUMBERS

Algorithm · A set of step-by-step instructions a computer follows.

Answer key / reference · The trusted measurement we compare against. For us, weather radar.

Baseline · The normal dry-weather level of a link's signal loss. Rain is measured above it.

Bias · A consistent lean in one direction. Our totals read about 18% low, so we have a low bias.

Calibration · Adjusting a measurement so it matches a trusted reference, like setting a kitchen scale to zero.

Correlation (r) · A 0-to-1 score of how closely two readings rise and fall together. 0.95 means almost in step.

Dashboard · A screen of live charts and numbers. Our Control Room is one.

Data pipeline · The chain of automatic steps that turns raw data into results.

Dataset · A collection of data. Ours: 500 links, 11 days, readings every minute.

Detection rate · Of the real events, the share we caught. Also called "hit rate" or POD (probability of detection).

False alarm rate · Of the alarms we raised, the share that were wrong.

Grid / cell · Splitting a map into equal squares. Each square is a cell.

Held-out data · Data kept aside and used only for the final test, never for tuning.

Interpolation · Estimating values between known points. How we fill the map between links.

Inverse distance weighting (IDW) · An interpolation method where nearer points count more.

Open-source · Software whose code is free for anyone to use and check. We used an open-source tool called pycomlink.

Prototype · A working early version that proves an idea, not the finished product.

Real-time · Updated live, as things happen. Our map refreshes every 10 minutes.

Resolution · How fine the detail is. "1 km, 10 minutes" means 1 km squares updated every 10 minutes.

Standard deviation / "wobble" · How much readings jump around. We use it to spot when rain starts.

Wet antenna effect · Extra loss from a film of water on the dish. We subtract it.

INSURANCE AND MONEY

Actuary · A maths expert who prices insurance.

Basis risk · The chance the trigger and the real loss don't match.

Cap · The maximum number of payouts allowed per year.

Claim · A request for payment with proof, in normal insurance. Kavach needs none.

Indemnity insurance · Normal insurance that pays your proven loss after checking.

Insurer / underwriter · The licensed company that carries the risk and pays out.

IRDAI · Insurance Regulatory and Development Authority of India, the regulator.

Loss ratio · Payouts as a share of premiums. We plan for about 60%.

Micro-insurance · Insurance with very small premiums and payouts, built for low-income people.

Parametric insurance · Pays automatically when a measured index (like rainfall) crosses a pre-agreed trigger.

Payout · The money paid when the trigger fires: ₹300 for Kavach.

PMFBY · Pradhan Mantri Fasal Bima Yojana, the government's crop insurance scheme.

Policy · The insurance agreement.

Premium · What you pay for cover: ₹29 a month for Kavach.

Regulatory sandbox · A trial zone run by a regulator where new products can be tested with limited customers.

Reinsurance · Insurance for insurance companies, so one huge storm can't bankrupt them.

Trigger · The pre-agreed rule that causes a payout, e.g. 10 mm of rain within 3 hours.

Waiting period · Days after buying before cover starts (7 for Kavach), to stop last-minute buying.

Business, strategy and policy words

BUSINESS AND STRATEGY

API · Application Programming Interface: a way for another company's computer system to plug into our data automatically.

B2C / B2B / B2G / B2B2C · Who you sell to: consumers, businesses, government, or a business that buys for its own people.

Business case · The argument for why an idea will make (or save) enough money to be worth doing.

CAC (customer acquisition cost) · What it costs to win one customer. Near zero for Kavach, thanks to the recharge.

Customer zero · When a company is the first user of its own product. Reliance's own businesses use JioMausam first.

Distribution · How a product reaches customers. Ours rides on Jio's recharge.

Distribution fee · What Jio earns for selling the insurance.

Data licence fee · What Jio earns for providing the trigger data.

Flywheel · A self-reinforcing cycle: more users make the product better, which brings more users.

Go-to-market (GTM) · The plan for launching and selling a product.

Gig worker · Someone paid per task through an app: delivery riders, cab drivers. NITI Aayog expects 23.5 million by 2029–30.

Informal sector · Work without formal contracts, salary slips or benefits. About 90% of India's workers.

Moat · What makes a business hard to copy.

Monetise · To turn something into money.

Pilot · A small real-world trial before full launch.

Platform · A base system other products are built on. JioMausam is the platform; Kavach is one product on it.

Roadmap · The planned steps and timeline.

Scalable · Can grow to many more users without costs growing as fast.

Unit economics · The money made or lost on one customer: here ₹348 premium, about ₹210 expected payout.

Value proposition · Why a customer should care. For Ravi: "₹29 a month, paid automatically when weather stops your work."

RELIANCE COMPANIES

Jio Platforms · Reliance's digital and telecom arm, including the Jio network and MyJio app.

JioPay / JioFinance · Reliance's payment and finance apps, our payout channel.

Jio Financial Services (JFS) · Reliance's listed financial services company, which works with insurance partners.

JioMart · Reliance's online grocery and quick-delivery business.

Reliance New Energy · Reliance's clean energy business (solar, batteries, hydrogen).

GOVERNMENT AND RULES

DPDP Act 2023 · Digital Personal Data Protection Act: India's law on how personal data is collected and used, with consent.

Consent · The user clearly agreeing to how their data is used.

Anonymised / aggregated data · Data with names removed and combined by area, so no individual can be identified.

State disaster management authority · The state body that handles floods, heatwaves and other emergencies. A likely buyer of our data.

Heat action plan · A city's plan to protect people during heatwaves. Ahmedabad was the first in South Asia.

Kavach Basic / Plus · Basic: bought by individuals, pays on the city's extreme days. Plus: bought by platforms for workers, pays on any work-stopping day.

GSOD / synoptic station · NOAA's Global Summary of the Day archives daily reports from official weather stations worldwide, including IMD's airport stations.

How to explain it to anyone

10 SECONDS

“Rain weakens the signals between mobile towers. We use that to map weather street by street, and pay workers automatically when it stops their work.”

30 SECONDS

“India has only about 40 weather radars, so nobody knows which streets got hit by a storm. But Jio’s towers talk over radio beams, and rain weakens those beams in a measurable way. JioMausam reads that to build a live rain map with no new hardware. On top, Mausam Kavach pays workers ₹300 automatically when extreme rain or heat hits their area: from ₹29 a month at recharge, priced city by city on 25 years of IMD data, or free to riders when their delivery app buys it. We tested it on 500 real links: it matches official radar at 0.95.”

2 minutes: the structure

1. **Hook:** “India’s biggest weather network already exists. It’s called Jio.”
2. **Problem:** weather is street-level, data isn’t; workers lose income; insurance is too slow.
3. **Insight:** rain weakens tower signals; phones feel heat. We become the sensor.
4. **Proof:** 500 real links, 0.95 match with radar, 91% of rainy days caught; priced on 25 years of IMD data.
5. **Product:** JioMausam map + Kavach (Basic from ₹29/month, Plus via delivery apps), ₹300 automatic payouts.
6. **Business:** premiums, data sales, Reliance as customer zero. ₹174 crore at 1% adoption.
7. **Why Reliance:** network, users, payments, insurance and businesses, all under one roof.
8. **Ask:** a 90-day Mumbai monsoon pilot (rain, riders, one delivery app), then heat, then everyone.

Explain it to your grandmother

“You know how the TV signal goes fuzzy when it rains hard? Mobile towers have the same problem. We use that fuzziness to tell exactly where it’s raining. Then, if a poor worker has paid ₹29 a month and it pours where they are, the phone company sends them ₹300 that same day, without any forms.”

If you forget everything else, remember these numbers

~40	weather radars in all of India
500M	Jio users (our reach)
500	real links we tested on
0.95	our match with radar (network-wide)
91%	rainy days caught (4.6% false alarms)
₹29-52 / ₹300	Basic price by city / payout
33%	of Mumbai heavy-rain days when a station 20 km away got under half the rain
₹174 Cr	yearly premiums if 1% of Jio users buy

Questions you might be nervous about

Short, simple answers. For many more, see the 199-question bank in Part 3.

“How do you know it actually works?”

We tested it on 500 real network links and compared with official weather radar. Network-wide our readings matched at 0.95 out of 1, and we caught 91% of rainy days. Scientists have shown the same thing in Israel, the Netherlands, Germany and Sweden.

“Why is your data from Germany?”

Telecom companies don't publish their data. The only open research dataset is German. The physics is the same in India, and our heavier monsoon rain gives an even clearer signal. Indian data is what the pilot is for.

“Doesn't Jio use fibre cables?”

Much of it, yes, and fibre doesn't sense rain. But many sites still use microwave links, and we add phone data plus official radar and gauges. Measuring real coverage is the pilot's first job.

“Is Jio becoming an insurance company?”

No. A licensed insurer carries the risk. Jio sells it at recharge and provides the weather data, earning fees for both.

“Why would a poor worker pay?”

₹29 a month is less than a cup of tea a week, it's added in one tap, and it pays without any forms. Often their delivery app would pay for them.

“What about privacy?”

Rain uses tower equipment data, not personal data. Heat uses only phones whose owners agree, combined by area, following the DPDP Act.

“What if it rains on someone and they don't get paid?”

That's called basis risk. We keep the squares small, combine several data sources, and offer a review option. Our test already agreed with radar on 94.6% of payout decisions, and the pilot will push that higher.

“Can't Airtel copy this?”

They could read rain, but they don't have payments, insurance, retail and energy businesses to use it. And whoever maps India first owns the best data.

“What will it cost Reliance?”

Mostly a small team and computing. No new towers or machines. The network was the expensive part, and it's already built.

“What's the biggest risk?”

Not having enough microwave links in some cities. That's why the pilot measures it first, and why we combine several data sources instead of relying on one.

Answers to expert feedback: the technology

We had the idea reviewed by an expert. These are the toughest technical points and how we answer them. Each one has a number the pilot will measure.

THE CHALLENGE	OUR ANSWER	THE PILOT MEASURES
Links miss 32% of heavy-rain events and read 18% low.	Multi-sensor fusion. Links are one input, never the only one. We blend them with IMD Doppler radar, municipal rain gauges and satellite, and calibrate every area against gauges continuously. Proven blending methods include gauge-adjusted merging and machine learning; fuzzy-logic blending is another option. We pick by measured accuracy, not fashion.	Heavy-rain events caught: 80% or more. Totals within $\pm 10\%$ of gauges.
Jio's fibre can't sense rain, so street-level density may not exist.	Phase 1 is a coverage audit. We count the usable microwave and E-band links in every 1 km square of the Mumbai grid and publish a coverage map. Where links are too sparse, that square leans on radar and gauges, or we don't sell cover there yet.	Usable links per square; share of the city with enough links.
A link measures a line, not a point.	True, and our prototype simplified it by placing each link's rain at its midpoint. Production uses path-aware interpolation, which treats each reading as the average along its line (research methods exist for exactly this), and gives squares covered mainly by long links a lower confidence score.	Map error per square against gauges, split by link length.

IN PLAIN WORDS

Think of each link as telling you the average rain along a road, not at one house. Many crossing roads, plus a few official gauges as anchors, let you rebuild what happened on each street.

SAY IT LIKE THIS (ANSWER · PROOF · BRIDGE)

"Links alone catch 68% of heavy-rain events, so we never use them alone. We fuse them with IMD radar and gauges, and our pilot target is 80%. That's the advantage of owning the densest sensor network in the city."

Answers to expert feedback: the money

Premiums are not Jio's revenue

EXAMPLE	PREMIUMS A YEAR	BACK TO WORKERS (60%)	JIO'S FEES (15–25%)
Kavach Plus: one app, 50,000 Mumbai riders	₹12.5 Cr	₹7.5 Cr	₹1.9–3.1 Cr
Kavach Basic: 1% of Jio's users	₹174 Cr	₹104 Cr	₹26–44 Cr

A licensed insurer holds the premiums and carries the risk. Jio earns distribution and data fees. Data sales and savings inside Reliance come on top and are Jio's own.

Stress test: the worst year in 25 (IMD records, 2000–2024)

PRODUCT	AVERAGE PAYOUT DAYS	PRICE	WORST YEAR	PAYOUTS ÷ PREMIUMS THAT YEAR	YEARS LOSING MONEY
Mumbai Plus (heavy rain, cap 5)	11.9, so 5 paid	₹208	2010: 19 days, 5 paid	60%	0 of 25
Mumbai Basic (204.5 mm+)	1.1	₹45	2019: 4 days	222%	7 of 25
Nagpur Basic (47°C+)	2.2, so 1.2 paid	₹52	2013: 14 days, 5 paid	242%	5 of 25
Ahmedabad Basic (47°C+)	0.7	₹29	2018: 7 days, 5 paid	469%	4 of 25

What the stress test tells us

- **Basic is real insurance.** Bad years cost 2–5 times the premiums, so it needs reinsurance and pooling across cities. Mumbai's rain and Nagpur's heat don't peak in the same months.
- **Correlated losses are bounded.** A city-wide flood day costs ₹1.5 crore per 50,000 riders, and the 5-day cap limits anyone's yearly payout to ₹1,500.
- **Mumbai Plus hits its cap every year.** For a platform it works as an automated rain allowance that lands on the right days for the right riders, rather than a gamble. A 3-day version costs about ₹125 a rider a month.

SAY IT LIKE THIS

“₹174 crore is the premium pool, not our revenue. Jio's share is 15 to 25 percent, and the insurer carries the risk. We stress-tested 25 years of IMD data: Basic's worst years need reinsurance; Plus never lost money.”

Why a delivery app pays, and how triggers get calibrated

The case for a delivery app: operational continuity

Predictable weather cost	Today a storm means surprise rain bonuses for riders and lost orders. Kavach turns that into a fixed premium the app can budget for.
Riders stay through the monsoon	A rider who is paid on a washed-out day stays attached to the app instead of switching apps or quitting. Replacing riders means recruiting, onboarding and training again.
Welfare and the law	The Code on Social Security, 2020 asks platforms to contribute 1–2% of turnover to gig-worker welfare, and states such as Rajasthan and Karnataka have their own gig-worker laws. Kavach is a benefit riders can actually feel.
The price is small per rider	₹208 a month is roughly 1% of a full-time rider's monthly earnings, or about ₹7 a day. A 3-day version costs about ₹125.

BE HONEST ABOUT THIS

We don't yet know the app's own costs: how many orders a storm loses, how many riders go offline, how churn moves after the monsoon. The shadow pilot measures exactly these in the partner app's data, so the ROI case is built on their numbers, not ours.

Daily history, 3-hour triggers: how they line up

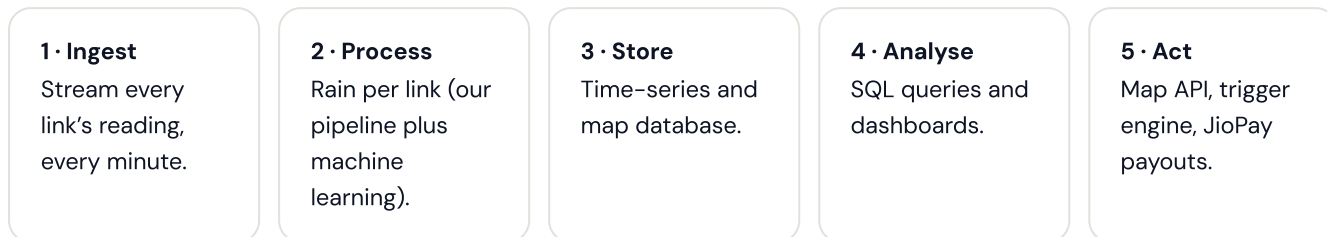
- 1 Today:** Indian weather history is daily, so our prices come from daily IMD records. The 5-day cap keeps any mismatch bounded.
- 2 During the shadow pilot:** we record real 3-hour rain from the network and Mumbai's automatic rain gauges, and count how often each 3-hour trigger level would fire.
- 3 Before money moves:** Jio Financial Services' actuaries recalibrate the 3-hour trigger to a target of about 4–5 payout days a year, and stress-test it against city-wide floods like 2005.

SAY IT LIKE THIS

“For a delivery app this is operational continuity: storm costs become a fixed premium and riders stay through the monsoon. The pilot measures the ROI in their own data before anyone signs a cheque.”

Answers to expert feedback: operations

The data pipeline at national scale



Lakhs of links × 1,440 readings a day is hundreds of millions of readings: a streaming job, not a script. SQL is the analysis layer on top, not the whole pipeline.

The rule for riders who move around

1. Riders are covered by the app's own **delivery zone** (about 2–5 km), not by a single 1 km square.
2. A zone triggers when its squares, averaged, cross the trigger within a 3-hour window.
3. A rider qualifies if logged in to the app for at least **60 minutes** of that window in that zone.
4. If a rider worked in several zones, the zone with the most logged-in time counts.
5. At most **one payout per rider per day**.

When the network itself fails in a storm

SITUATION	WHAT THE SYSTEM DOES
Some links in a square are down	Uses the remaining links and neighbouring squares; that square's confidence score drops.
All links down during heavy rain	Treats the outage itself as a rain signal and switches that square to IMD radar and automatic weather stations.
Radar also unavailable	Uses satellite rainfall and municipal gauges; the decision may wait up to 24 hours.
Still can't decide	Pays if a neighbouring square within 2 km triggered, and reconciles afterwards.

THE RULE TO REMEMBER

“An outage can delay a payout. It can never cancel one.”

Answers to expert feedback: regulation and the pilot

The 90-day pilot validates; it doesn't launch

For 90 days we run everything except the money. The system computes every payout it **would** make ("shadow mode"), but no insurance is sold and nobody is paid. We compare those decisions with radar, gauges and the delivery app's own data, and share the results with Reliance and the partner.

WHEN	WHAT HAPPENS
Months 0–3	Coverage audit and shadow pilot. No product is sold, so it needs no product approval.
Months 3–6	Apply to IRDAI's regulatory sandbox with a Jio Financial Services insurance partner.
Months 4–12	Sandbox: real payouts to a capped group of one app's riders. Heat module built separately.
Year 2	File the full product. Launch Kavach Basic at recharge.

The pilot's go / no-go scorecard

Heavy-rain events caught (after fusion)	80% or more
False triggers	under 15%
Rain totals compared with gauges	within ±10%
Simulated payout time	under 6 hours
Share of Mumbai with enough usable links	measured and published (it sets where we can sell)

Unproven parts are kept separate

Heat sensing from phones is phase 2 research, and selling to individuals at recharge is phase 3. Neither is part of the pilot's claims, so they can't drag down confidence in the rain platform.

THE THREE HARDEST CHALLENGES, IN ONE BREATH EACH

"Fibre? Phase 1 is a coverage audit that counts our usable links." "Basis risk? Fusion with radar and gauges, target 80%, measured in shadow mode before money moves." "Revenue? ₹174 crore is the premium pool; Jio's fees are 15 to 25 percent, with no insurance risk."

How to drill, together

The answer formula: A · P · B

Answer

One sentence that answers exactly what was asked.

Proof

One number or fact. Not three.

Bridge

One line back to a strength. Then stop talking.

EXAMPLE

“Doesn’t Jio mostly use fibre?” A: “Much of it, yes, and fibre doesn’t sense rain.” P: “But many sites still use microwave, and we fuse in radar and gauges.” B: “Measuring real coverage is the first thing our 90-day pilot does.”

Five drills

1. **Flashcards (20 min):** one reads a question from the bank, the other answers aloud in under 30 seconds, then reads the written answer. Swap every 10 questions.
2. **Rapid-fire (5 min):** 10 random question numbers in 3 minutes, no pause longer than 3 seconds.
3. **Hostile judge (10 min):** one plays a sceptical judge and interrupts. Stay calm, use A · P · B, never argue.
4. **Numbers drill (2 min):** recite the numbers on the “How to explain it” page without looking.
5. **Full run (12 min):** video, deck against a timer, then 4 minutes of Q&A from the bank. Record it on a phone and watch it once.

Split the questions

Teammate 1	Categories 2, 3, 4, 8: technology, data, India, scaling.
Teammate 2	Categories 5, 6, 7, 10: product, insurance, business, regulation.
Both	Categories 1, 9, 11–14. Whoever owns a topic answers; the other adds one line only if something important was missed.

Four-day plan

Day 1	Read Parts 1 and 2 together. Do the self-test at the end.
Day 2	Flashcards on categories 1–7.
Day 3	Flashcards on categories 8–14, then rapid-fire across all of them.
Day 4	Two full timed runs. Fix the three answers you stumbled on.
Your slot	Morning: numbers drill and tech checklist. Then stop practising and rest.

GOLDEN RULES

Never bluff: say “we haven’t measured that yet; the pilot will”. Never argue with a judge. Keep every answer under 30 seconds.

199 cross-questions, with answers

Grouped by topic. Every answer is written to be said aloud in under 30 seconds, and every number matches our prototype and IMD analysis. Use the Q numbers for rapid-fire drills.

1 • The idea itself

What judges ask first. Answer each in one breath.

Q1 What is JioMausam, in one sentence?

Software that turns Jio's network into a street-level weather sensor, plus Mausam Kavach, which uses those readings to pay workers automatically when extreme rain or heat stops their work.

Q2 What problem are you solving?

Outdoor and gig workers lose income to extreme weather with no protection. Nobody measures weather street by street, and checking a ₹300 claim costs more than paying it.

Q3 What makes it 'Different by Design'?

Everyone else adds sensors: radars, stations, satellites. We add none. The design choice is to reuse a network that already exists in every district.

Q4 Who is the customer?

Three groups: workers (Kavach Basic), the platforms that employ gig workers (Kavach Plus), and insurers, states and businesses that buy the data. Reliance's own businesses are customer zero.

Q5 Who uses it and who pays?

The worker always benefits. The payer is the worker for Basic, the delivery app for Plus, or an employer or state scheme.

Q6 Why now?

Heat and cloudbursts are getting worse, gig work is booming (7.7 million workers in 2020–21, 23.5 million expected by 2029–30), UPI makes ₹300 transfers instant and free, IRDAI has a sandbox, and Jio reaches 500 million people. All of that only came together recently.

Q7 What's the one thing we should remember?

The network is already a weather sensor. We proved it on 500 real links, priced it on 25 years of Indian data, and it needs no new hardware.

Q8 Is this a telecom, weather or insurance product?

A data platform with an insurance product on top. Telecom is the sensor; insurance is the first and most visible use.

Q9 What does success look like in three years?

A national 1 km weather map refreshed every 10 minutes, crores of covered workers, and JioMausam data used by insurers, states and Reliance's own businesses.

Q10 What's your focus?

Rain first, riders first, Mumbai first. Rain is the most proven part, delivery apps paying for riders is the most realistic way to sell, and Mumbai has the most heavy-rain days to test on.

Q11 What are your phases?

1: a Mumbai coverage audit with payouts in shadow mode (rain, riders, one delivery app). 2: real payouts in IRDAI's sandbox, plus heatwaves in Nagpur and Ahmedabad. 3: Kavach Basic at every recharge. 4: a national map and data API.

Q12 Why rain before heat?

Rain sensing from links is proven at national scale. Heat from phone batteries is published research but not yet run as a service. We lead with what's proven.

Q13 Why riders before individuals?

Low-income workers rarely buy insurance on their own. One deal with a delivery app covers thousands of riders at once, free to them, which builds trust.

Q14 What's futuristic about it?

Networks that sense their surroundings ('integrated sensing and communication') are a 6G research goal. We do a practical version today with 4G/5G data, and our insurance settles itself from live data.

2 • How the technology works

The science questions. Use the torch-in-the-rain picture, then add one technical detail.

Q15 How exactly does rain affect a microwave link?

At 6–40 GHz, raindrops absorb and scatter the beam. The extra loss grows with rain rate following a standard formula (ITU-R P.838): loss per km = $a \times \text{rain rate}^b$.

Q16 Why does frequency matter?

Higher frequencies feel rain more. Links between about 15 and 40 GHz give a strong signal; links below about 10 GHz barely react.

Q17 Why does link length matter?

Loss builds up along the path, so longer links give a bigger signal, but they average the rain over a longer line.

Q18 What data do you need from the network?

Transmitted and received signal levels for each link every 1–15 minutes, plus each link's frequency, length, polarisation and tower locations.

Q19 Do operators really record this?

Yes. It's standard performance data that network operations centres use to watch link health. We reuse it.

Q20 What is the wet antenna effect?

A film of water on the dish adds loss even in drizzle. We subtract a modelled amount (Schleiss 2013 method) so we don't overcount rain.

Q21 How do you tell rain from an equipment fault?

Rain makes the signal wobble and drop gradually. We flag 'wet' when a link's one-hour variability is unusually high for that link. Faults look different (sudden steps, flat lines) and are filtered out.

Q22 What's the baseline?

The link's normal dry-weather loss, carried forward through a rain spell. Rain is only the loss above it.

Q23 How do you turn links into a map?

We estimate each grid square from the nearest links, nearer ones counting more (inverse distance weighting). In production we'd fuse links with radar and gauges using ML.

Q24 What resolution can you reach?

In cities, about 1 km every 10 minutes. In rural areas it's coarser, and every square carries a confidence score.

Q25 How does heat sensing work?

Phone battery temperature follows the air around it. Averaged over many consented phones in an area, it tracks air temperature (shown on 8 cities, 2013). Tower-site temperature sensors add fixed points, and IMD stations calibrate it.

Q26 Don't pockets and charging spoil battery readings?

For one phone, yes. Averaging thousands and filtering out charging or heavy-use phones removes most of it. That's why heat is area-level, never individual.

Q27 Could it measure anything else?

Fog and humidity also affect high-frequency links. Water vapour, fog for highways and airports, and air quality are future modules.

Q28 What does 5G add?

5G radio-quality data also reacts to rain, and E-band backhaul links (70–80 GHz) are very rain-sensitive. Denser 5G means a denser sensor.

Q29 What if Jio uses fibre?

Fibre doesn't sense rain. But microwave remains at many sites, especially rural, hilly and fast-rollout ones. The pilot measures real coverage first, and we fuse with radar, gauges and satellite.

Q30 What software did you use?

pycomlink, the open-source research library from the Karlsruhe Institute of Technology, plus our own calibration, testing, India analysis and two web prototypes.

Q31 How 'live' is it?

Links log every 1–15 minutes, processing takes seconds, the map refreshes every 10 minutes, and payouts reach wallets within hours.

Q32 Could traffic load be mistaken for rain?

No. A fixed link's received power doesn't depend on how much data it carries. Interference and multipath look different and are filtered by the wet/dry step.

Q33 How much computing does national scale need?

Lakhs of links at one reading a minute is a few hundred million readings a day. That's a modest streaming job for Reliance's data centres.

Q34 Is the physics settled?

Yes. The rain-microwave relationship has been engineering standard for decades (it's why satellite TV breaks up in storms). Using it for rainfall maps has been peer-reviewed since 2006.

3 • Data, accuracy and the prototype

Know these numbers cold. Say where they came from.

Q35 What data did your prototype use?

500 commercial microwave links, 1-minute signal levels, 10–20 May 2018: the public OpenSense/pycomlink research sample, with German Weather Service radar as the reference.

Q36 Why German data?

No Indian operator publishes link data. This is the best open dataset in the world. The physics is identical, and Indian monsoon rain is heavier, which makes the signal stronger.

Q37 Are the locations real?

It's a real network; the provider shifted the coordinates for privacy. Distances and geometry are preserved.

Q38 How did you measure accuracy?

We compared our rain with radar, along each link's path and on a 4 km grid, hour by hour and day by day.

Q39 What does a correlation of 0.95 mean?

Network-average rain from our method and from radar rose and fell almost identically, hour by hour, on days the model had never seen.

Q40 And for a single link?

Median 0.88. 91% of links scored above 0.7.

Q41 What are 'held-out' days and why do they matter?

We tuned one number on days 1–3 and tested on days 4–11. Testing on unseen days proves we didn't just fit the answer.

Q42 What did you calibrate?

One multiplying factor, $\times 1.17$, because the method read slightly low. Nothing else was tuned to the test days.

Q43 Your totals read 18% low. Why?

The simple baseline and wet-antenna steps undercount light rain. It's a known effect, fixed by regular calibration against rain gauges.

Q44 Give me the detection numbers.

Rainy days: 91% caught, 4.6% false alarms. Wet hours (1 mm/h or more): 61% caught, 15.7% false alarms. Heavy days (10 mm or more): 62% caught, 14% false alarms.

Q45 What is the 94.6% payout agreement?

For about 14,000 square-days, the pay/no-pay decision using our data matched the decision using radar 94.6% of the time.

Q46 Isn't 94.6% inflated by dry days?

Partly, yes, and we say so. The tougher number: we caught 68% of radar's trigger events, and 18% of our triggers weren't confirmed by radar. Fusing in gauges and radar is how we close that gap.

Q47 Is radar the truth?

No. Radar has its own errors (beam height, blocking, its own attenuation). It's the best independent reference available. The pilot adds rain gauges.

Q48 How would you improve accuracy?

More links (density), several frequencies, machine-learning wet/dry detection, and fusion with gauges, radar and satellite, calibrated locally.

Q49 What happened in the Kavach simulation?

1,000 simulated policyholders, the real storm's rain at their locations, and the 10 mm in 3 hours rule: 445 were paid ₹1,33,500 in total, a median of about 70 minutes after their rain started.

Q50 Did you write the code?

The signal processing uses the open-source pycomlink library. The pipeline, calibration, testing, India pricing, the Kavach engine and both prototypes were built for this project.

Q51 Can we see it working?

Yes: the Control Room (five tabs, including the India layer) and the Mausam Kavach phone app both open in any browser.

Q52 What's the weakest part of your evidence?

No Indian tower data yet. That's exactly what the 90-day pilot fixes, and it's why our first ask is access to Jio's link logs for Mumbai.

4 • India-specific questions

Our new evidence. These show the idea fits Indian reality.

Q53 Where does your Indian data come from?

IMD station reports for Mumbai Santacruz, Mumbai Colaba, Nagpur and Ahmedabad, 2000–2024, from NOAA's global archive of weather-station reports (the Global Summary of the Day).

Q54 Why not straight from IMD?

Our build environment couldn't reach IMD's site, and the same station reports are archived by NOAA. In the pilot we'd use IMD's gridded and automatic-weather-station data directly.

Q55 How reliable is that archive for India?

Good for temperatures and for counting rain days. It can under-record single extreme days: it shows 26 July 2005 at Santacruz as 461 mm against IMD's official 944 mm. So we use it to count how often thresholds are crossed, not for exact totals.

Q56 Is Indian rain really hyperlocal?

Yes. On 26 July 2005 Santacruz recorded 944 mm and Colaba, 20 km away, 73 mm. And on 1 in 3 heavy-rain monsoon days (2019–24), one Mumbai station got less than half the other's rain.

Q57 How often does Mumbai get heavy rain?

About 11.9 days a year of 64.5 mm or more (IMD 'heavy'), 4.4 days of 115.6 mm or more, and 1.1 days of 204.5 mm or more ('extremely heavy').

Q58 And heat?

Nagpur: about 9.7 days a year at 45°C or hotter and 2.2 at 47°C. Ahmedabad: 5.4 at 45°C and 0.7 at 47°C. Mumbai: no days at 44°C or more.

Q59 So what should Kavach cost?

Kavach Basic covers each city's extreme days: Mumbai ₹45 a month (204.5 mm days), Nagpur ₹52 (47°C days), Ahmedabad ₹29 (47°C days, costing ₹27). Kavach Plus covers any work-stopping day, up to 5 a year: about ₹208 a rider a month in Mumbai.

Q60 Why did your price change from a flat ₹29?

Indian data showed one price can't fit every city: Mumbai has about 12 heavy-rain days a year. We kept ₹29 as the floor and priced each city on its own history, which is what an actuary would do.

Q61 Why Mumbai for the pilot?

It has the most triggers to test on: about 12 heavy-rain days a year, against about 2.4 in Nagpur, so one monsoon gives a dozen events to measure accuracy. It's one of India's biggest delivery-rider markets, and our best Indian evidence is from Mumbai.

Q62 What about cyclones, river floods or hail?

Cyclones bring heavy rain, which the rain trigger covers. River floods need hydrology models (a later module). Hail is hard to detect from links and remains a research area.

Q63 Does India have enough microwave links?

Indian networks have used a lot of microwave backhaul, especially in rural areas, though fibre is growing. The exact count is internal to Jio, which is why measuring it is the pilot's first deliverable.

Q64 What about language and literacy?

The app already works in Hindi and would add regional languages, plus WhatsApp and voice alerts. Payouts need no action at all.

5 • The product and the customer

Questions about Mausam Kavach from the worker's side.

Q65 How does a worker buy it?

One tap while recharging in MyJio or at a retailer. KYC is already done for the SIM. Cover starts 7 days later.

Q66 What exactly triggers a payout?

Basic: the city's extreme threshold in your ~1 km square (for example 47°C in Ahmedabad, or 204.5 mm in a day in Mumbai). Plus: any work-stopping day, 10 mm of rain in 3 hours or 45°C heat, up to 5 days a year.

Q67 How much, and how fast?

₹300 per trigger day into JioPay, usually within hours. In the prototype it was about 70 minutes after the rain started.

Q68 Why ₹300?

It's roughly a third to half of a day's earnings for many outdoor workers: enough to matter, small enough to keep premiums affordable.

Q69 Why not pay a full day's wage?

The premium would roughly triple. Partial cover keeps it affordable, and platforms can buy higher cover for their workers.

Q70 How do you know where the worker is?

With consent, the network knows which area their phone is in. We only use the ~1 km square, never an exact location.

Q71 Why would someone in an air-conditioned office buy it?

They wouldn't, and that's by design. Kavach is for people whose income stops when the weather turns. Office workers get free street-level alerts in MyJio instead, and their employers can buy the data.

Q72 Who checks that buyers really work outdoors?

For Plus, the delivery app, which knows its riders. For Basic, e-Shram registration (the government's register of unorganised workers) or occupation details at sign-up. The price covers expected payouts either way.

Q73 What if they move around the city?

Cover follows the phone's area during working hours. For riders, the platform can share the zones they worked.

Q74 What if they don't have a smartphone?

It works with any phone: SMS alerts and UPI payouts. Jio's JioBharat feature phones support UPI.

Q75 Do they have to do anything after buying?

No. They just get a message with the reading that triggered the payout.

Q76 What if they disagree with a decision?

They tap 'Review my day'. We check gauges and radar, and pay if they confirm the trigger. Disputes also improve the model.

Q77 Why would people renew?

Visible payouts, a tiny price tied to the recharge, and their own language. Platforms renew on behalf of riders.

Q78 What does the app show day to day?

Live rain or heat at your location, progress towards the trigger, alerts, payout history, and the plan with its city price.

Q79 How do you win the trust of first-time insurance users?

Platforms pay first, so it's free to the worker. The first payout arrives without asking. Every payout shows its reading. And it comes from a brand they already use.

Q80 How do you stop people buying only before a storm?

A 7-day waiting period, monthly auto-renewal with the recharge, and seasonal pricing.

Q81 Can farmers buy it?

Farm workers and small farmers can take seasonal monsoon or heat plans. Crop-loss cover is a later module with insurers.

6 • Insurance and pricing

Where judges test whether you understand the money side of risk.

Q82 What is parametric insurance?

Insurance that pays when a measured index crosses a pre-agreed trigger, instead of after you prove a loss.

Q83 Why are you paying customers? Isn't Reliance giving money away?

No. Customers pool small premiums, and the insurer pays the few who have a bad day from that pot. In Ahmedabad, 1 lakh customers pay ₹3.48 crore a year, payouts are about ₹1.92 crore, and the rest covers the insurer, reinsurance and Jio's fees.

Q84 Who carries the risk?

A licensed insurer through Jio Financial Services' partners, backed by reinsurance. Jio is distributor and data provider.

Q85 How did you price it?

Average trigger days a year from 25 years of IMD data, capped at 5, $\times ₹300$, divided by a 60% target loss ratio, divided by 12 months.

Q86 Walk me through Ahmedabad.

47°C or hotter happens about 0.7 days a year (0.64 after the cap). $0.64 \times ₹300 = ₹192$ of expected payouts. $\div 0.6 = ₹320$ a year, about ₹27 a month. We charge ₹29.

Q87 Why a 60% loss ratio?

It leaves 40% for the insurer's costs and capital, reinsurance, distribution and data fees. That's achievable because there's no claims handling.

Q88 What is basis risk and how big is yours?

The gap between the trigger and the real loss. Our prototype makes the same decision as radar 94.6% of the time and catches 68% of trigger events. Fusion improves both.

Q89 Why cap payouts at 5 days a year?

It keeps the premium affordable and predictable. Extreme seasons are covered by reinsurance.

Q90 What happens in a 2005-type disaster year?

Many triggers at once. The per-person cap limits payouts, and reinsurance pays for the extreme tail.

Q91 Climate change will raise payouts. Then what?

We reprice every year for each area, weighting recent decades more. And the data itself becomes more valuable as extremes grow.

Q92 Why would an insurer partner with you?

Cheap distribution to 500 million people, an independent trusted trigger, and near-zero claims cost: the three things micro-insurers lack.

Q93 Will IRDAI accept a network-based trigger?

Weather-index insurance already exists in India. Our trigger source is new, so we start in IRDAI's regulatory sandbox and validate against IMD.

Q94 How is this different from crop insurance schemes?

Government crop schemes use district or block-level data and pay farmers for crop losses weeks later. We use street-level data and pay workers for lost days within hours.

Q95 Isn't ₹208 a month expensive for Mumbai riders?

It's about ₹7 a day of monsoon-season cover, paid by the platform, not the rider. Platforms already spend on rider insurance, and gig-worker welfare laws push in this direction.

Q96 Could people buy cover for just one day?

Possibly later ('₹5 for today'), but that invites buying only on forecast storm days, so it would be priced on the forecast.

Q97 What if the network fails during a storm?

The reading falls back to radar and gauges. A total link outage in heavy rain is itself a strong rain signal.

7 • Business model and unit economics

Have the arithmetic ready. Say 'illustrative' for projections.

Q98 How does Jio make money from this?

A distribution fee on premiums, a data licence fee from the insurer, API sales of weather data, and savings inside JioMart, New Energy and Jio.

Q99 Show me the ₹174 crore.

500 million users $\times 1\% = 5$ million policies. $\times ₹29 \times 12 = ₹174$ crore of premiums a year. About 60% goes back as payouts.

Q100 What's Jio's share of that?

Illustratively 15–25% as distribution and data fees: ₹26–44 crore a year at 1% adoption from Basic alone, before Plus and data sales.

Q101 What are the costs?

A small data team, cloud computing, partnerships and customer messaging. No hardware. Payouts are carried by the insurer.

Q102 What does a customer cost to acquire?

Close to nothing for Basic, because it's an add-on inside the recharge. Plus needs B2B sales to platforms.

Q103 What's a customer worth over time?

Illustratively, a ₹29-a-month customer kept for two years pays about ₹700 in premiums. Jio's fee share is ₹100–175, plus the value of keeping them on Jio.

Q104 What does Plus look like as a business?

₹208 a rider a month in Mumbai. A platform with 50,000 riders pays about ₹12.5 crore of premiums a year; Jio's fees on that are about ₹1.9–3.1 crore.

Q105 Who pays for the data?

Insurers pricing weather products, state disaster agencies, municipal flood cells, farm-input and logistics companies, and renewable energy forecasters.

Q106 What's 'customer zero' worth?

JioMart: rain shifts quick-commerce demand by the hour. New Energy: forecast errors cost grid penalties. Jio: rain-fade outages. Small percentage gains in each are worth crores.

Q107 Is the market big enough?

India has hundreds of millions of outdoor and informal workers. Even 1% of Jio's base is 5 million policies, before B2B and data.

Q108 Why not just sell the data and skip insurance?

Insurance creates visible value for people, plus brand and loyalty. Data alone is a smaller B2B market. We do both.

Q109 How does it help Jio's core telecom business?

Kavach customers have a reason to stay, so churn falls. It adds revenue per user through the add-on, and it improves network planning.

Q110 What happens in year one?

The pilot year earns little; it proves accuracy and payouts. Year two adds ten cities, the first platform deals and the first data contracts.

Q111 Which metrics would you track?

Trigger accuracy against gauges, payout time, policies and renewals, payouts as a share of premiums, platform deals and data revenue.

Q112 What if individuals don't buy?

Lead with platforms paying for workers, state schemes funding cover for informal workers, and data sales. The business doesn't depend on individuals alone.

8 • Scaling and operations

How it grows from two districts to India.

Q113 How do you get from Mumbai to all of India?

The software scales with the network. Each new city needs data access, calibration against IMD stations, and trigger prices from its weather history.

Q114 What's hardest about scaling?

Getting clean data across equipment makers and regions, building the fusion model, and regulatory approval for each product.

Q115 How many people does it need?

About 5 for the pilot and 30–50 nationally: data science, network engineering, product, partnerships and compliance.

Q116 How do you handle different equipment vendors?

Standardise their logs into one format (the research community already has an open one, CMLH5) inside Jio's data platform.

Q117 How do you calibrate nationally?

Continuous comparison with IMD radar, automatic weather stations and rain gauges, with models tuned for each region.

Q118 What about rural areas with few towers?

Coarser resolution, flagged with a confidence score, with radar and satellite filling in. Rural areas often rely more on microwave links, which helps.

Q119 Can it go beyond India?

Yes: monsoon countries with few radars and dense mobile networks, such as Bangladesh, Indonesia and much of Africa. The method can be licensed to other telecoms.

Q120 Which partners do you need?

IMD for validation, insurers through JFS, gig platforms, state disaster authorities, and research groups (KIT, SMHI) for a fast start.

Q121 How long to national coverage?

About two years for major cities and three for national coverage.

Q122 What's the production technology?

Streaming ingestion of network data, processing (pycomlink-style plus ML), a geospatial store, APIs, and payout integration with JioPay and UPI.

Q123 How do you keep it running 24×7?

Redundant processing, automatic fallback to radar and gauges, and monitoring. If a payout is delayed, it's reconciled later.

Q124 What could slow you down?

Internal approvals for data access, regulatory timelines and partner contracts. That's why the ask is a small pilot with clear criteria.

9 • Competition and moat

Why nobody else can do this, and why it can't easily be copied.

Q125 Who are your competitors?

IMD and private weather firms on data, insurers and insurtechs on parametric cover, Tomorrow.io globally, and other telecoms.

Q126 Why can't a private weather company do this?

They don't have an Indian operator's network data or a 500-million-person distribution channel.

Q127 Why can't Airtel copy it?

They could sense rain, but they don't have payments, insurance, retail and energy businesses under one roof. And whoever maps India first owns the reference data.

Q128 What about satellites?

Satellite rainfall is coarse (about 10 km, every 30 minutes) and sees cloud tops, not street-level rain. We fuse it; we don't compete with it.

Q129 Tomorrow.io started this way. Why didn't they win India?

They moved to satellites and global models, and never had Indian operator data or Indian distribution.

Q130 What exactly is your moat?

Exclusive network data, plus distribution, payments, insurance and internal users, all in one company. And a flywheel: more users make better maps, which make fairer payouts, which bring more users.

Q131 Can the science be patented?

The core science is public. The moat is exclusive data, integration and distribution, plus our own calibration and fusion models.

Q132 What stops insurers going around Jio?

They need the trigger data and the distribution. Both are Jio's.

Q133 Could the government do this itself?

IMD could partner with us, and we'd welcome that. But IMD doesn't run a telecom network or sell insurance.

Q134 What if a startup partners with Airtel?

Possible, which is why speed matters. Reliance can move first and go deeper, because it owns every layer.

10 • Regulation, privacy and ethics

Calm, specific answers. Name the law or regulator.

Q135 Is using network data legal?

The link performance data is Jio's own equipment data. Aggregated weather outputs aren't personal data.

Q136 How do you comply with the DPDP Act 2023?

Personal data (location, phone sensors) is used only with explicit consent, for a stated purpose, kept to a ~1 km area, and can be withdrawn at any time.

Q137 Do TRAI or DoT need to approve?

Using network data for internal analytics generally doesn't need approval. We'd still review licence conditions with DoT and legal teams.

Q138 Can a telecom sell insurance?

Yes, as a corporate agent or distribution partner under IRDAI rules, or through JFS's licensed entities.

Q139 What is the IRDAI regulatory sandbox?

A framework where new insurance products are tested with a limited number of customers for a limited time before full approval.

Q140 Is it fair if some areas get better data than others?

We publish coverage and confidence for every area, set triggers with data quality in mind, and fall back to other sources where coverage is thin.

Q141 Could this be used for surveillance?

The rain method uses no personal data. Heat uses only consented, aggregated data, under strict governance.

Q142 Is it ethical to sell insurance to the poor?

It's cheap, transparent, automatic and often paid for by platforms. The alternative today is no protection at all.

Q143 How do you prevent mis-selling?

A simple product, a clear trigger, cancel anytime, and no claims process to trip people up.

Q144 Where is the data stored?

In India, on Reliance infrastructure.

Q145 What if you pay out by mistake?

The cost is small and built into pricing. We never claw money back from customers.

Q146 Doesn't this compete with IMD?

No. It complements IMD by adding density, and we'd share data for public warnings.

11 • Risks and failure modes

Show you've thought about failure before they ask.

Q147 Honestly, how realistic is this?

The science is proven, weather insurance already exists in India, and our one real unknown, link coverage, is exactly what the 90-day Mumbai pilot measures first. The data platform is realistic in 1–2 years; insurance through delivery apps in 2–3.

Q148 What's your biggest risk?

Coverage: too few microwave links in some cities. The pilot measures it first, and fusion reduces our dependence on any one source.

Q149 What if accuracy in India is worse than in Germany?

Heavier rain helps. If needed, we add gauge and radar fusion and set more conservative triggers.

Q150 What if no insurer partners with you?

Start with Plus under a platform's own group policy, or with state-funded schemes for informal workers.

Q151 What if people don't trust automatic payouts?

The first payout builds trust. Every payout shows its reading, and platforms fund the first season.

Q152 What if there's a huge disaster year?

Reinsurance, per-person caps and yearly repricing.

Q153 What if Jio's network team won't share data?

They gain from it too (predicting rain fade). We start with two districts and anonymised outputs.

Q154 What about fraud?

Parametric cover removes most fraud: nobody can make it rain on a tower. Location checks stop gaming.

Q155 What if equipment changes break the model?

Continuous calibration and monitoring for every link.

Q156 What if regulations change?

The product is simple and transparent, and starting in the sandbox keeps us close to the regulator.

Q157 What would make you abandon the idea?

If the pilot showed that, even with fusion, accuracy in Indian cities can't beat existing gauges and radar. Then we'd pivot to a data-fusion product without network sensing.

12 • Reliance strategy and fit

Connect the idea to Reliance's own businesses.

Q158 How does this fit Reliance's strategy?

Jio's push into AI and data services, JFS's insurance ambitions, Retail's quick commerce and New Energy's solar forecasting can all run on one platform.

Q159 Which Reliance company should own it?

Jio Platforms (data and AI), with Jio Financial Services as the insurance partner and a cross-company steering group.

Q160 How does it help New Energy?

Site-level solar forecasts, fewer grid deviation penalties and better scheduling.

Q161 How does it help JioMart?

Hourly demand forecasts tied to local rain, for better staffing and stocking in quick commerce.

Q162 How does it help Jio's network?

Predicting rain fade, planning link capacity and prioritising maintenance.

Q163 How does it help India?

Climate adaptation, financial inclusion and disaster early warning, complementing Mission Mausam.

Q164 What's the sustainability angle?

Climate resilience for the most vulnerable people, with measurable social impact for Reliance's ESG reporting.

Q165 Is this part of Jio's AI story?

Yes. Weather is a natural AI product: fusion models, rain nowcasting and risk scores.

Q166 Why should Reliance trust students with this?

We built a working prototype on real data in days, and our ask is a small, measurable pilot.

Q167 What would you ask a Reliance mentor first?

How to start the link-log access process, and who the product owner in JFS would be.

13 • Your team and execution

Personalise these with your own story.

Q168 What did you build yourselves?

The rain pipeline on 500 links, calibration and testing, the India pricing on 25 years of IMD data, the Control Room, the phone app, and the business model.

Q169 How long did it take?

Days, using open-source tools, which shows how fast a real team inside Reliance could move.

Q170 What skills are you missing?

Actuarial pricing and telecom operations, which Reliance and JFS already have.

Q171 What would you do in the pilot's first 30 days?

Get data access, inventory the links, map coverage, measure accuracy against IMD, and sign partner MoUs.

Q172 How will you measure the pilot?

Correlation of 0.8 or more against gauges, at least 80% of heavy-rain events caught, fewer than 15% false triggers, and payouts within 6 hours.

Q173 Why are you the right team?

Tell your own story: what you learned building it, and why the problem matters to you (for example, seeing workers lose days to rain).

Q174 What was hardest?

Admitting the flat ₹29 didn't survive Indian data, and redesigning the product around city-by-city pricing.

Q175 What would you do differently?

Approach an operator for Indian tower data earlier, and build the heat module in parallel.

14 • Curveballs and rapid-fire

Short, confident answers. Smile.

Q176 If you had ₹10 crore, where would it go?

The team and data platform first, then the insurance sandbox and customer education in two cities, then the data API.

Q177 And if you had only ₹10 lakh?

A one-district coverage study using Jio's logs, plus ten low-cost rain gauges to validate it.

Q178 Explain it to a 10-year-old.

Rain makes phone-tower signals weaker. We use that to see where it's raining and send money to people who can't work because of it.

Q179 What's your idea's weakest point?

It depends on how many microwave links exist in each city. We're open about that, and the pilot tests it first.

Q180 Why should we pick you over other teams?

A working prototype on real data, Indian pricing from 25 years of records, honesty about limits, and an idea only Reliance can execute.

Q181 This sounds too good to be true.

That's why we tested it on days it had never seen (0.95 against radar) and why our ask is a 90-day pilot with success criteria, not a big cheque.

Q182 What does the name mean?

Mausam means weather. Kavach means shield.

Q183 Which single number matters most?

Zero: the hardware we need to add. Or 0.95: how closely we match radar.

Q184 What if you don't win?

We'd publish the open prototype and still approach Jio, JFS, a state government or a platform for a pilot.

Q185 What comes after rain and heat?

Flood risk, fog for roads and airports, air quality, and crop risk.

15 • Expert follow-ups

The hardest questions from our expert review. Answer, prove, bridge.

Q186 How will you close the gap where links miss 32% of heavy-rain events?

We never use links alone. We fuse them with IMD radar, municipal rain gauges and satellite, calibrated per area. The pilot target is 80% of heavy-rain events caught.

Q187 How do you fix the 18% underestimate?

Continuous calibration against rain gauges plus fusion. The pilot target is totals within $\pm 10\%$ of gauges.

Q188 Links measure a line, not a point. Doesn't that distort your map?

Yes, and our prototype simplified it by placing each link's rain at its midpoint. Production uses path-aware interpolation that treats each reading as a line average, and gives squares covered by long links lower confidence.

Q189 Which fusion method? Fuzzy logic?

Several proven options: gauge-adjusted merging, kriging with radar as a guide, machine learning; fuzzy-logic blending is one more. We'll choose by measured accuracy in the pilot, not by fashion.

Q190 What if Mumbai turns out to have too few microwave links?

That's why Phase 1 is a coverage audit. Squares with too few links lean on radar and gauges, or we don't sell cover there yet.

Q191 Is ₹174 crore your revenue?

No. It's the premium pool; a licensed insurer holds it. Jio's fees are 15–25%: ₹26–44 crore at 1% adoption. For one app's 50,000 riders, ₹12.5 crore of premiums means ₹1.9–3.1 crore of fees for Jio. Data sales are on top.

Q192 Why would a delivery app pay ₹208 a rider a month?

It turns unpredictable storm costs (rain bonuses, lost orders) into a fixed premium, keeps riders attached through the monsoon, and delivers a visible welfare benefit as social-security rules for gig workers tighten. The shadow pilot measures the ROI in the app's own data.

Q193 Your prices use daily data but Plus pays on 3-hour readings. How do you reconcile them?

Indian history is daily. The shadow pilot records real 3-hour frequencies from the network and Mumbai's automatic gauges, and JFS actuaries recalibrate the trigger to about 4–5 payout days a year before any money moves.

Q194 What happens when a city-wide flood triggers everyone at once?

It's bounded. One flood day costs ₹1.5 crore per 50,000 riders, and the 5-day cap limits anyone's yearly payout to ₹1,500. Reinsurance and pooling across cities cover bad years.

Q195 Isn't Mumbai Plus just a guaranteed payment, since it always hits the cap?

In 25 years of data it paid its cap every year, so for platforms it works as an automated rain allowance that lands on the right days for the right riders. A 3-day version costs about ₹125 a month. Basic, by contrast, is real insurance.

Q196 How do you handle riders moving across zones in a 3-hour window?

Cover is by the app's delivery zone, not one square. A zone triggers when its squares, averaged, cross the trigger; a rider qualifies with 60 minutes logged in there; the zone with the most logged-in time counts; one payout per day at most.

Q197 What if the network fails during the storm itself?

We fall back in order: remaining links and neighbouring squares, then IMD radar and weather stations, then satellite and municipal gauges. An outage can delay a payout, never cancel one.

Q198 How do you process hundreds of millions of readings a day?

A streaming pipeline: ingest every link reading, compute rain per link, store it in a time-series and map database, and analyse it with SQL and dashboards. SQL is the analysis layer, not the whole pipeline.

Q199 Is the 90-day pilot a commercial launch?

No. It's a technical and product validation: we compute every payout in shadow mode but send no money. Real payouts start in IRDAI's sandbox in phase 2.

Twelve-question self-test

Cover the answers, answer out loud, then check. If you get 8 or more right, you're ready for the jury.

1. In one sentence, what does JioMausam do?
2. What happens to a microwave link's signal when it rains, and why?
3. What is a "baseline", and why do we need it?
4. What is the difference between normal insurance and parametric insurance?
5. What are Kavach's two plans, their prices and triggers?
6. What does a correlation of 0.95 mean?
7. Why did we test on "held-out days"?
8. Why is 94.6% agreement not the whole story, and what's the tougher number?
9. Name the five puzzle pieces only Reliance holds.
10. What's the biggest risk, and how does the pilot handle it?
11. What did 25 years of IMD data tell you?
12. What's your focus, and what are the four phases?

Answers

1. It uses the weakening of mobile-tower radio signals (and phone temperatures) to map weather street by street, and powers automatic insurance payouts.
2. It gets weaker, because raindrops soak up and scatter the beam. More rain, more weakening.
3. The link's normal dry-day signal loss. Rain is only the extra loss above it, so without the baseline we'd count normal loss as rain.
4. Normal insurance pays a proven loss after checking a claim. Parametric pays automatically when a measured event crosses a pre-agreed trigger.
5. Basic: ₹29–52 a month by city, ₹300 on the city's extreme days (e.g. 47°C in Ahmedabad). Plus: bought by delivery apps, ₹300 on any work-stopping day such as 10 mm of rain in 3 hours, up to 5 days a year.
6. Our readings and the radar's rise and fall almost perfectly together.
7. To prove the method works on data it was never tuned on, like an exam with unseen questions.
8. Most days are dry, so "no payout" is easy to get right. The tougher number: we caught 68% of the radar's heavy-rain events.
9. Sense (network), reach (500M users), pay (JioPay), insure (JFS and partners), use (JioMart, New Energy, Jio).
10. Too few microwave links in some areas (much of the network is fibre). The pilot measures real coverage first, and we combine several data sources.
11. Rain is hyperlocal (on a third of Mumbai's heavy days, a station 20 km away got under half the rain), and each city has a different danger: Mumbai rain, Nagpur and Ahmedabad heat. So one flat price can't work.
12. Rain first, riders first, Mumbai first. Phases: 1 Mumbai monsoon pilot with one delivery app; 2 heatwaves in Nagpur and Ahmedabad; 3 Kavach Basic at every recharge; 4 a national map.

YOU'RE READY WHEN

You can tell Ravi's story, explain the torch-in-the-rain idea, and say "0.95 against radar on 500 real links" without looking at notes.